
Notes on Optimization under Uncertainty

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1. Introduction

The core problem: Standard deterministic optimization assumes all model parameters are known, but in reality, some key parameters like demand or prices are random, and we cannot optimize a random objective function directly.

Three ways to deal with the uncertainty:

1. Mean-value heuristic:

$$\max_{x \geq 0} F(x, \mathbb{E}^{\mathbb{P}}[D]), \quad (1)$$

but maybe suboptimal, that is, *the flaw of averages*¹;

2. Robust optimization (very conservative): maximize the worst-case profit over $D \in [l, u]$, that is

$$\max_{x \geq 0} \min_{D \in [l, u]} F(x, D); \quad (2)$$

3. Stochastic programming: treat D as a proper random variable and

$$\max_{x \geq 0} \mathbb{E}^{\mathbb{P}}[F(x, D)]. \quad (3)$$

The decision-maker makes an optimal decision when the realized values of random parameters are unknown but their probability distributions are known (or can be estimated).

2. Stochastic Programming Modeling

Unknown decision variable values but the probability distributions are known.

The essential departure from deterministic optimization:

$$\underbrace{\max_x F(x, \mathbb{E}^{\mathbb{P}}[D])}_{\text{deterministic: plug in mean}} \longrightarrow \underbrace{\max_x \mathbb{E}^{\mathbb{P}}[F(x, D)]}_{\text{stochastic: optimize expectation}} \quad (4)$$

General Stochastic Program (SP):

$$\min_{x \in \mathcal{X}} \mathbb{E}_{\xi} [F(x, \xi)] = \min_{x \in \mathcal{X}} \int_{\Omega} F(x, \xi(\omega)) d\mathbb{P} \quad (5)$$

where ξ is a random vector defined on some probability space $(\Omega, \Sigma, \mathbb{P})$ and $\mathcal{X}$ is a given set of deterministic constraints.

To model a general stochastic program, we first examine the natural timing of real decisions, specifically, which decisions must be committed to before uncertainty resolves and which can adapt afterward. This timing structure leads us to formulate the problem as a two-stage or multi-stage stochastic program². Once the model is formulated, we can then solve it using methods such as the extensive form, Benders decomposition, or sample average approximation.

2.1 Temporal structure of decision-making

Two-stage: (1) First stage (here-and-now): decisions made before uncertainty is revealed. (2) Second stage (recourse): adaptive decisions made after observing the random outcome.

¹The flaw of averages: the optimal response to an average scenario is not the average optimal response. The mathematical root is *Jensen's inequality*.

²The stages split comes from the problem itself, not from mathematical convenience.

The Two-stage structure is introduced to capture the information timeline in real problems:

$$\underbrace{\text{commit to } x}_{\xi \text{ unknown}} \longrightarrow \underbrace{\xi \text{ is revealed}}_{\text{uncertainty resolves}} \longrightarrow \underbrace{\text{choose recourse } y}_{\xi \text{ now known}} \quad (6)$$

and it is a faithful model of the reality that some decisions must be made before uncertainty resolves, and others can adapt afterward.

Distinguish: what decision must be committed to before uncertainty resolves (first stage), what decision can wait until after uncertainty is revealed (second stage), and the clear information revelation event between the two stages (the uncertainty realization point).

Multi-stage: The general version is multi-stage SP:

$$\text{decision 1} \longrightarrow \xi_1 \longrightarrow \text{decision 2} \longrightarrow \xi_2 \longrightarrow \dots \quad (7)$$

But multi-stage SP suffers from exponential scenario tree growth, so in practice, two-stage is preferred whenever it is a reasonable approximation of the true decision timeline.

2.2 Two-Stage SP

The general two-stage stochastic program is:

$$\min_{x \in \mathcal{X}} c^\top x + \mathbb{E}_\xi[Q(x, \xi)], \quad (8)$$

where the recourse function $Q(x, \xi)$ is defined by a second optimization:

$$Q(x, \xi) := \min q(\xi)^\top y \quad \text{s.t.} \quad W(\xi)y = h(\xi) - T(\xi)x, \quad y \geq 0 \quad (9)$$

where x is the first-stage decision and the most important output of model, and y is the recourse decisions, implicitly depend on ξ . Three objects can be random: the recourse cost $q(\xi)$, the $T(\xi)$, and the right-hand side $h(\xi)$.

Two core challenges: Computing $\mathbb{E}_\xi[Q(x, \xi)]$ is a potentially high-dimensional integral $Q(x, \xi)$ has no closed form, and it is defined implicitly via an optimization.

Extensive Form When ξ takes finitely many values $\{\xi^1, \dots, \xi^S\}$ with probabilities p_s , the problem becomes a single large linear programming (LP):

$$\begin{aligned} \min_{x, y^s} \quad & c^\top x + \sum_{s=1}^S p_s (q^s)^\top y^s \\ \text{s.t.} \quad & Ax = b, \\ & W^s y^s = h^s - T^s x, \quad \forall s, \\ & x \geq 0, \quad y^s \geq 0, \quad \forall s. \end{aligned} \quad (10)$$

When the uncertainty has a finite number of scenarios, or when the original stochastic problem is approximated by a finite set of scenarios, the SP can be reformulated as an equivalent extensive-form LP. The construction starts from the deterministic model corresponding to a nominal scenario. We then distinguish between first-stage decisions, which must be chosen before the uncertainty is revealed, and second-stage decisions, which are made after the realization of uncertainty is known. Since the second-stage decisions depend on the realized scenario, each recourse variable is assigned a scenario index s . In the objective function, the second-stage costs are weighted by the scenario probabilities p_s , so that the model minimizes the expected total cost. Finally, the second-stage constraints are imposed separately for each scenario. This produces a single large LP in which the first-stage variables are shared across all scenarios, while the second-stage variables are scenario-specific.

Note: The key structural observation: x is shared across all scenarios (the linking variable), while y^s are scenario-specific. This block-angular structure motivates *Bender's decomposition*.

Recourse Type	Condition	Meaning
Complete recourse	$Q(x, \xi) < \infty \forall x \in \mathbb{R}^n, \xi \in \Xi$	Second stage always feasible, even infeasible x
Relatively complete recourse	$Q(x, \xi) < \infty \forall x \in \mathcal{X}, \xi \in \Xi$	Second stage feasible for all feasible first-stage x
Fixed recourse	$W(\xi) = W$ for all ξ	Recourse matrix is deterministic
Simple recourse	$W = [I, -I]$	Each constraint has its own surplus/slack

Recourse Problem: Recourse refers to the opportunity to adapt a solution to specific outcome observed. If your first-stage decision x turns out to pair badly with the realized outcome ω , the recourse variable y gives a chance to recover.

In the second stage, after uncertainty is revealed, given a first-stage decision x committed to

$$Q(x, \xi) := \min q(\xi)^\top y \quad \text{s.t.} \quad W(\xi)y = h(\xi) - T(\xi)x, \quad y \geq 0. \quad (11)$$

y is the corrective action, which adapts to whatever scenario ξ actually occurred, trying to repair any mismatch between the first-stage commitment x and the realized uncertainty. The total objective balances the upfront first-stage cost $c^\top x$ against the expected cost of these future corrections $\mathbb{E}[Q(x, \xi)]$.

The central concern is **feasibility**: if no recourse action y can fix the situation for some scenario $Q(x, \xi) = +\infty$ and the model breaks down. Good practice therefore demands *relatively complete recourse*, ensuring that for every feasible first-stage decision and every possible scenario, a finite-cost recourse always exists. The standard way to guarantee this is to introduce *penalized slack variables* that absorb any infeasibility at a finite but high cost, rather than letting the model become mathematically undefined.

Two ways: Penalize Shortfall and Chance Constraints Original Optimization Problem:

$$\begin{aligned}
\min \quad & \sum_{i \in \mathcal{I}} f_i x_i + \sum_{s \in \mathcal{S}} p_s \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} c_{ij} y_{ij}^s \\
\text{s.t.} \quad & \sum_{i \in \mathcal{I}} y_{ij}^s \geq d_j^s, & \forall j \in \mathcal{J}, s \in \mathcal{S}, \\
& \sum_{j \in \mathcal{J}} y_{ij}^s - u_i x_i \leq 0, & \forall i \in \mathcal{I}, s \in \mathcal{S}, \\
& y_{ij}^s \geq 0, & \forall i \in \mathcal{I}, j \in \mathcal{J}, s \in \mathcal{S}, \\
& x_i \in \{0, 1\}, & \forall i \in \mathcal{I}.
\end{aligned} \quad (12)$$

1. **Penalize Shortfall** Instead of requiring demand to always be met, allow unmet demand but charge a penalty

$$\begin{aligned}
\min_{x_i, y_{ij}^s, z_j^s} \quad & \sum_{i \in \mathcal{I}} f_i x_i + \sum_{s \in \mathcal{S}} p_s \left[\sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} c_{ij} y_{ij}^s + \sum_{j \in \mathcal{J}} \lambda_j z_j^s \right] \\
\text{s.t.} \quad & \sum_{i \in \mathcal{I}} y_{ij}^s + z_j^s \geq d_j^s, & \forall j \in \mathcal{J}, s \in \mathcal{S}, \\
& \sum_{j \in \mathcal{J}} y_{ij}^s - u_i x_i \leq 0, & \forall i \in \mathcal{I}, s \in \mathcal{S}, \\
& z_j^s \geq 0, & \forall j \in \mathcal{J}, s \in \mathcal{S}, \\
& y_{ij}^s \geq 0, & \forall i \in \mathcal{I}, j \in \mathcal{J}, s \in \mathcal{S}, \\
& x_i \in \{0, 1\}, & \forall i \in \mathcal{I}.
\end{aligned} \quad (13)$$

where $\sum_{j \in \mathcal{J}} \lambda_j z_j^s$ is the expected penalty for unmet demand in the second stage and z_j^s absorbs unmet demand so $Q(x, \xi^s) < +\infty$.

2. Chance Constraints Require the second-stage constraints to hold in at least $1 - \alpha$ fraction of scenarios:

$$\begin{aligned}
\min_{x, y^s} \quad & c^\top x + \sum_{s=1}^S p_s (q^s)^\top y^s \\
\text{s.t.} \quad & Ax = b, \\
& W^s y^s = h^s - T^s x, \quad \forall s \in \mathcal{S}_\alpha, \\
& \sum_{s=1}^S p_s \cdot \mathbf{1}_{[s \in \mathcal{S}_\alpha]} \geq 1 - \alpha, \\
& x \geq 0, \quad y^s \geq 0, \quad \forall s.
\end{aligned} \tag{14}$$

Or equivalently, with binary variables $v^s \in \{0, 1\}$ indicating whether demand is met in scenario s :

$$\begin{aligned}
\min_{x, y^s, v^s} \quad & c^\top x + \sum_{s=1}^S p_s (q^s)^\top y^s \\
\text{s.t.} \quad & Ax = b, \\
& W^s y^s \geq h^s - T^s x - M v^s, \quad \forall s, \\
& \sum_{s=1}^S p_s v^s \leq \alpha, \\
& x \geq 0, \quad y^s \geq 0, \quad v^s \in \{0, 1\}, \quad \forall s.
\end{aligned} \tag{15}$$

where $v^s = 1$ means scenario s is allowed to violate feasibility, and M is a large constant. This turns the problem into a mixed-integer program.

VSS and EVPI Using the minimization framing here:

1. Value of the Stochastic Solution (VSS):

$$\nu^S := \min_{x \in \mathcal{X}} \mathbb{E}_\xi [F(x, \xi)] \tag{16}$$

2. Expected Value of Perfect Information (EVPI): Solve using $\mathbb{E}[\xi]$ as if deterministic, get x^{MV} , then evaluate its true cost:

$$\nu^{MV} := \mathbb{E}_\xi [F(x^{MV}, \xi)] \tag{17}$$

3. Perfect information value (wait-and-see):

$$\nu^{PI} := \mathbb{E}_\xi \left[\min_{x \in \mathcal{X}} F(x, \xi) \right] \tag{18}$$

Then:

$$\text{VSS} := \nu^{MV} - \nu^S \geq 0 \quad (\text{cost of ignoring uncertainty}) \tag{19}$$

$$\text{EVPI} := \nu^S - \nu^{PI} \geq 0 \quad (\text{value of a perfect oracle}) \tag{20}$$

The rank is always:

$$\nu^{PI} \leq \nu^S \leq \nu^{MV} \tag{21}$$

Note: for maximization problems the signs flip:

$$\text{VSS} = \nu^S - \nu^{MV}, \quad \text{EVPI} = \nu^{PI} - \nu^S. \tag{22}$$

Example: CVaR Assume finite support distribution:

$$\mathbb{P}(\xi = \xi^s) = p_s, \quad s = 1, \dots, S. \quad (23)$$

for a cost random variable Z and confidence level α :

$$\text{CVaR}_\alpha(Z) = \min_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{1-\alpha} \mathbb{E}[Z - \eta]_+ \right\} \quad (24)$$

Intuitive interpretation: when Z is continuous:

$$\text{CVaR}_\alpha(Z) = \mathbb{E}[Z \mid Z \geq \text{VaR}_\alpha(Z)] \quad (25)$$

i.e. the expected cost in the worst $(1 - \alpha)$ fraction of scenarios. Setting $\alpha = 0.95$ means you are optimizing the average over the worst 5% of outcomes.

$$\min_{x, \eta} \left\{ \eta + \frac{1}{1-\alpha} \mathbb{E}[f(x, \xi) - \eta]_+ : x \in \mathcal{X}, \eta \in \mathbb{R} \right\}, \quad (26)$$

that is,

$$\begin{aligned} \min_{x, \eta, w} \quad & \eta + \frac{1}{1-\alpha} \sum_{s=1}^S p_s w_s \\ \text{s.t.} \quad & x \in \mathcal{X}, \\ & \eta \in \mathbb{R}, \\ & w_s \geq f(x, \xi^s) - \eta, \quad \forall s = 1, \dots, S, \\ & w_s \geq 0, \quad \forall s = 1, \dots, S. \end{aligned} \quad (27)$$

Convex program when $\mathcal{X}$ is convex and $f(x, \xi^s)$ is convex for each s .

In practice one often uses a convex combination:

$$\min_x (1 - \lambda) \mathbb{E}[F(x, \xi)] + \lambda \text{CVaR}_\alpha(F(x, \xi)) \quad (28)$$

where $\lambda \in [0, 1]$ controls the trade-off between average performance and tail protection.

2.3 Multi-Stage SP

In the multi-stage, sequential decisions across 3 or more than 3 stages, uncertainty unfolds gradually, and decisions x_t are required to be non-anticipative (depend only on information $\xi_{[t]} = (\xi_1, \dots, \xi_t)^3$ observed so far, not future outcomes).

Nested Formulation:

$$\min_{x_1 \in X_1} f_1(x_1) + \mathbb{E} \left[\inf_{x_2 \in X_2(x_1, \xi_2)} f_2(x_2, \xi_2) + \mathbb{E} \left[\dots + \mathbb{E} \left[\inf_{x_T \in X_T(x_{T-1}, \xi_T)} f_T(x_T, \xi_T) \right] \right] \right] \quad (29)$$

where

$$\begin{aligned} f_t(x_t, \xi_t) &:= c_t^\top x_t, \\ X_1 &:= \{x_1 : A_1 x_1 = b_1, x_1 \geq 0\}, \\ X_t &:= \{x_t : B_t x_{t-1} + A_t x_t = b_t, x_t \geq 0\}, \quad t = 2, \dots, T. \end{aligned} \quad (30)$$

Here

$$\begin{aligned} \xi_1 &:= (c_1, A_1, b_1), \quad \text{is known non-random, while} \\ \xi_t &:= (c_t, B_t, A_t, b_t) \in \mathbb{R}^{d_t}, \quad t \geq 2 \quad \text{may be partly or fully random.} \end{aligned} \quad (31)$$

In general, it's impossible to compute nested expectations over continuous random variables in closed form for most realistic problems.

³ $\xi_{[t]}$ has a specified probability distribution.

Scenario tree: It is a discrete representation of the stochastic process used to represent history of decision-making. A path from the root to a leaf is called a *scenario*. The nodes represent the states of information, and the edges are the realizations of uncertainty. The root node $f_1(x_1)$ is deterministic. Branching probability is assigned to branches, and the conditional probability is defined as moving from one node to the next. Scenario trees are used to define non-anticipativity constraints.

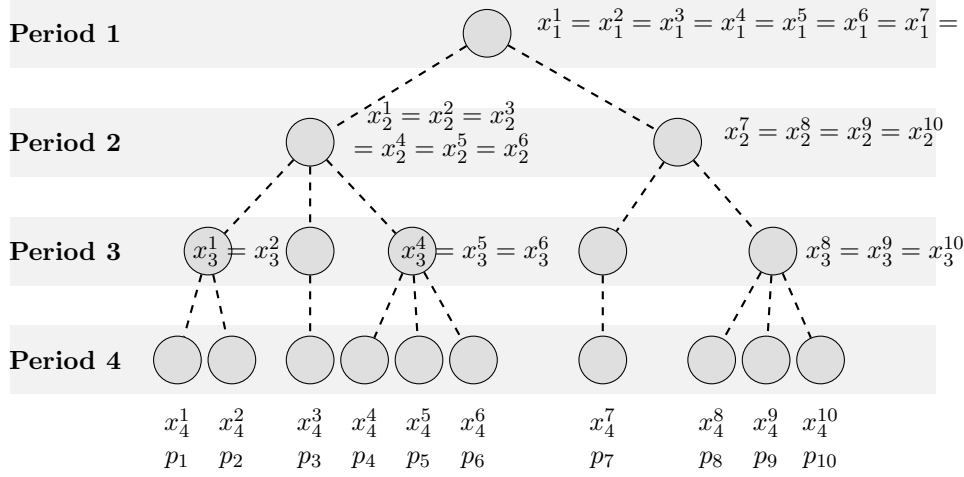


Figure 1: x_t^k represent decisions made for scenario k at time t .

Multi-stage scenario-based LP formulation Approximate ξ by a finite set of K scenarios (or nodes $s \in \mathcal{S}$) with probabilities p_k , arranged in a tree with branching structure.

$$\begin{aligned}
 & \min \sum_{k=1}^K p_k \left[c_1^\top x_1^k + (c_2^k)^\top x_2^k + \cdots + (c_T^k)^\top x_T^k \right] \\
 \text{s.t.} \quad & A_1 x_1^k = b_1, \quad B_t^k x_{t-1}^k + A_t^k x_t^k = b_t^k, \quad x_t^k \geq 0 \\
 & x_t^k = x_t^j \quad \forall k, j \text{ s.t. } \xi_{[t]}^k = \xi_{[t]}^j \quad (\text{explicit non-anticipativity})
 \end{aligned} \tag{32}$$

The general formulation:

$$\begin{aligned}
 & \max \sum_{s \in \mathcal{S}} p_s \cdot f(x^{(s)}, \xi^{(s)}) \\
 \text{s.t.} \quad & x_1^{(s)} = x_1, \quad \forall s \in \mathcal{S} \\
 & x_t^{(s)} = x_t(\bar{\xi}), \quad \forall t = 2, \dots, T, \quad \bar{\xi}_{t-1} \in \Xi_{t-1}, s \in \mathcal{S} : \xi_{[t-1]}^{(s)} = \bar{\xi}_{t-1} \\
 & x_t^{(s)} \in \mathcal{C}_t(\xi_{[t-1]}^{(s)}), \quad \forall t = 1, \dots, T, \quad s \in \mathcal{S}
 \end{aligned} \tag{33}$$

where in the feasible set $\mathcal{C}_t(\xi_{[t-1]}^{(s)})$, $x_t^{(s)}$ depends on the realization $\xi_{[t-1]}^{(s)}$ and the hidden dependence of $x_t^{(s)}$ is on $x_{[t]}^{(s)}$.

Scenario generation approaches: optimization, simulation-clustering, and hybrid.

Optimization-Based Methods: moment matching (at t):

$$\begin{aligned}
\min_{x,p} \quad & \sum_{i=1}^n \sum_{k=1}^4 w_{ik} (m_{ik} - M_{ik})^2 \\
\text{s.t.} \quad & \sum_j p_{tj} = 1, \quad p \geq 0, \\
& m_{i1} = \sum_j x_{ij} p_{tj} \quad (\text{1st moment: mean}), \\
& m_{i2} = \sum_j (x_{ij} - m_{i1})^2 p_{tj} \quad (\text{2nd moment: variance}), \\
& m_{i3} = \sum_j (x_{ij} - m_{i1})^3 p_{tj} \quad (\text{3rd moment: skewness}), \\
& m_{i4} = \sum_j (x_{ij} - m_{i1})^4 p_{tj} \quad (\text{4th moment: kurtosis}).
\end{aligned} \tag{34}$$

- Decision variables: both the realizations x (node values) and the probabilities p . The optimizer chooses both simultaneously so that the tree's empirical moments m_{ik} match the target/estimated moments M_{ik} as closely as possible.
- Sequential variant: solve this moment-matching problem one node at a time, moving stage by stage.
- Overall variant: solve one large joint optimization across all nodes in the tree simultaneously. More globally consistent, more expensive.

Simulation-Clustering Methods simulate many sample paths from the stochastic process, then cluster-reduce them into a smaller, tractable tree structure.

- Sequential variant: simulate and cluster stage by stage and build the tree incrementally, conditioning each stage's simulation on the already-fixed parent node.
- Parallel variant ("at root"): simulate entire paths from root to final stage all at once, then cluster the complete paths afterward into the tree in one step.

Hybrid Methods Combine both families, e.g., use simulation to generate candidate scenarios, then apply an optimization step (such as moment-matching) to refine or select branch values/probabilities. These methods aim to balance the statistical rigor of optimization with the computational practicality of simulation.

Sample Average Approximation (SAA) is one practical way to actually build a scenario tree when you don't have an exact discretization of the underlying stochastic process, and instead, you generate the tree by sampling. It belongs to how to build a sequence (each node) in simulation clustering. A lightweight, specific instance within the sequential simulation family, but without the clustering/reduction step.

Steps:

1. Build the branching structure first. Before sampling any values, decide the shape of the tree: how many branches emanate from each node at each stage.
2. More branches in early stages than later ones. The tree typically has a higher branching factor near the root and fewer branches as it moves toward later stages, since resolving uncertainty finely close-in-time matters more, while the tree would otherwise explode in size.
3. Same number of branches per node within a given stage. Rather than varying the branching factor node-by-node, a common simplification is to fix the branching number for all nodes within the same stage (e.g., every node in Stage 2 has exactly 2 children, every node in Stage 3 has exactly 3 children, etc.).

Even though SAA is conceptually simple, generating trees this way still faces the same exponential blow-up in CPU time as branching depth and stage count increase. Sampling doesn't avoid the curse of

dimensionality; it just gives you a mechanical way to populate the tree once its shape is fixed.

Example: Multi-stage mean-variance optimization: The portfolio consists of n risky assets whose returns are uncertain and evolve stochastically over the planning horizon T (Gülpınar et al., 2003). At $t = 0$, it is initial allocation. The problem is to decide the holding across multiple discrete time points $t = 1, \dots, T-1$ by rebalancing, buying or selling assets in response to how returns and risk have evolved. At $t = T$, the portfolio is cashed out. It is the classical Markowitz mean-variance trade-off (maximize the wealth and minimize risk), but extended to a multi-period, stochastic setting.

$$\begin{aligned}
\max \quad & \{\alpha \mathbb{E}[W_T] - (1 - \alpha) \text{Var}[W_T]\} && \text{mean-variance} \\
\text{s.t.} \quad & \sum_{i=1}^n w_{\text{root}}^i = 1 && \text{initial balance} \\
& \mathbf{r}_e \otimes \mathbf{w}_{a(e)} + \mathbf{x}_e = \mathbf{w}_e, \quad \forall e \in \mathbb{N} && \text{transaction at } e \\
& \sum_{i=1}^n x_e^i = h_t, \quad \forall e \in \mathbb{N} && \text{balance.}
\end{aligned} \tag{35}$$

where $\alpha \in [0, 1]$ is the risk-aversion parameter, $a(e)$ is the ancestor/parent node of e in the scenario tree used to encode non-anticipativity implicitly, $\mathbf{r}_e$ is the vector of realized asset returns at node e , $\mathbf{x}_e$ is the net trading decision at the node e (buy/sell adjustments), and $\mathbf{w}_e$ is the resulting portfolio weight at the node e after applying returns and trades. The transaction constraint ensures the new portfolio position $\mathbf{w}_e$ equals the old position grown by realized returns, plus whatever the investor decides to trade at this node. The balance constraint is the self-financing condition, which means the total net trading activity sums to some target/holding level h_t (often zero net cash flow, meaning any buying is funded by selling, with no external cash injection or withdrawal).

To solve the above problem, we need to unpack $\mathbf{x}_e$ into two separate decision vectors buy (b) and sell s , and each is with its own transaction cost rate.

$$\begin{aligned}
p + (1 - c_b)b_0 - (1 + c_s)s_0 &= w_0, \\
\mathbf{1}'b_0 - \mathbf{1}'s_0 &= 1 - \mathbf{1}'p.
\end{aligned} \tag{36}$$

where p is the the current portfolio position, b_0 and s_0 are the initial buy and sell decision vectors, and c_b and c_s are the unit transaction cost rates for buying and selling respectively. At time t , the realized return of asset j is $r_t^j(\rho^t)$, which depends on the history of stochastic data ρ^t (non-anticipativity idea). w_{t-1}^j is the prior hoding in asset j . The asset-by-asset version of the abstract node-ba transaction constraints are

$$\begin{aligned}
w_t^j &= r_t^j(\rho^t)w_{t-1}^j + (1 - c_b^j)b_t^j - (1 + c_s^j)s_t^j, \quad \forall j = 1, \dots, n, \quad t = 1, \dots, T-1, \\
w_T^j &= r_T^j(\rho^t)w_{T-1}^j.
\end{aligned} \tag{37}$$

The explicit version of the abstract balance constraint is

$$\mathbf{1}'b_t - \mathbf{1}'s_t = 0, \quad \text{for } t = 1, \dots, T-1. \tag{38}$$

which means total buying must exactly equal total selling.

N_t is the set of nodes at time t in the scenario trees. Each node $e \in N_t$ represents one possible discretized outcome of the stochastic process ρ^t up to time t . The scenario tree is literally the discretization of the continuous random path ρ^t into a finite set of representative events.

$$W_t = \sum_{e \in N_t} P_e \left(\hat{r}'_e (w_{a(e)} - \bar{w}_{a(e)}) \right), \tag{39}$$

where $\bar{w}_{a(e)}$ is the benchmark's weights at that same parent node (market benchmark), $\hat{r}'_e$ is $\mathbb{E}(r_t(\rho^t)|\rho^{t-1})$, the expected return for event e , conditional on the history up to the parent, and P_e is the probability of

event e . The objective is how much extra wealth the investor expect to get purely from deviating from the benchmark, weighted by the probability of each possible path through the tree.

Risk is defined as the variance of portfolio return relative to the benchmark,

$$\begin{aligned} \text{Var} [r_t(\rho_t | \rho^{t-1})' (w_{t-1} - \bar{w}_{t-1})] \\ = \sum_{e \in N_t} P_e (w_{a(e)} - \bar{w}_{a(e)})' (\Lambda + \hat{r}_e \hat{r}_e') (w_{a(e)} - \bar{w}_{a(e)}) - W_t^2. \end{aligned} \quad (40)$$

The LHS is the definition: the the variance of the realized active return (return relative to benchmark), where $r_t(\rho_t | \rho^{t-1})$ is the actual random return (not just its expectation), conditional on history up to $t-1$. The RHS is the calculation using the scenario tree, where Λ is the covariance matrix of the n asset returns.

In this problem, there are two sources of uncertainty

- Uncertainty due to variability of return at a given realization
- Uncertainty due to the discrete distribution of realizations, that is, various paths through the scenario tree

Combined with all the decomposition, we can get a single, well-defined **multistage stochastic quadratic program**:

$$\begin{aligned} \min_{w, b, s} \quad & \sum_{t=1}^T \alpha_t \sum_{e \in N_t} P_e \left[(w_{a(e)} - \bar{w}_{a(e)})' (\Lambda + \hat{r}_e \hat{r}_e') (w_{a(e)} - \bar{w}_{a(e)}) \right] \\ \text{subject to} \quad & p + (1 - c_b)b_0 - (1 + c_s)s_0 = w_0, \\ & \mathbf{1}' b_0 - \mathbf{1}' s_0 = 1 - \mathbf{1}' p, \\ & r_e \circ w_{a(e)} + (1 - c_b) \circ b_e - (1 - c_s) \circ s_e = w_e, \quad e \in N_I, \\ & \mathbf{1}' b_e - \mathbf{1}' s_e = 0, \quad e \in N_I, \\ & \sum_{e \in N_T} P_e (\hat{r}_e' (w_{a(e)} - \bar{w}_{a(e)})) \geq W, \\ & w_e^L \leq w_e \leq w_e^U, \quad e \in N, \\ & 0 \leq b_e \leq b_e^U, \quad e \in N_I \cup \{0\}, \\ & 0 \leq s_e \leq s_e^U, \quad e \in N_I \cup \{0\}. \end{aligned} \quad (41)$$

which is the variance formula, summed across all stages, each weighted by a stage-specific risk-aversion coefficient α_t . In this model, the objective is minimizing total risk (variance relative to benchmark) across the entire investment horizon, rather than directly maximizing the mean-variance objective $\max\{\alpha \mathbb{E}[W_T] - (1 - \alpha) \text{var}[W_T]\}$. The expected wealth part of the trade-off has been moved into a constraint instead of the objective. In portfolio optimization, instead of jointly maximizing return and minimizing variance in one objective, it would be better to minimize risk subject to a minimum required return. Besides, there are new more realistic position constraints (e.g., no short-selling beyond some bound or diversification/concentration limits via w_e^L, w_e^U) and trading volume limits (you can't buy/sell more than some maximum b_e^U, s_e^U per period, and trading is naturally non-negative).

Worst-Case Optimal Robust Decisions In reality, the forecasts are inherently inaccurate, and return and risk estimates always carry estimation error, and small errors here can produce badly performing optimal portfolios, so we can extend the optimal portfolio simultaneously with the worst case. Instead of seeking optimality in the average sense (as in traditional mean-variance models), we can strive to maintain the best possible performance under the most adverse plausible scenarios; in other words, it sacrifices some average return to secure a safeguard against downside risk. Based on that, in the worst-case modeling, given that only first-stage decisions constitute immediate decisions in the true sense with subsequent stages allowing for rebalancing or adjustment, the robust design focuses on ensuring investor survival regardless of the first-period return outcome. Besides, rather than relying on a single covariance matrix Λ , the model considers multiple competing risk measures within a ro-

bust design framework. The objective changed from $\max\{\alpha\mathbb{E}[W_T] - (1 - \alpha)\text{var}[W_T]\}$ to the worst-case $\max\min_{i,j}\{\alpha\mathbb{E}[W_T^i] - (1 - \alpha)\text{var}[W_T^j]\}$, that is

$$\min_w \left\{ \gamma \sum_{t=1}^T \alpha_t \sum_{e \in N_t} \max_i \left[P_e (w_{a(e)} - \bar{w}_{a(e)})^\top (\Lambda_i + r_e^\top r_e) (w_{a(e)} - \bar{w}_{a(e)}) \right] - \min_k \left[\sum_{e \in N_T^k} P_e (w_{a(e)} - \bar{w}_{a(e)})^\top r_e \right] \right\}. \quad (42)$$

where α_t decides how to allocate risk weighting across different stages, while γ is how much the whole risk budget matters relative to profit target overall. There are two additional constraints, the first one is

$$\sum_{e \in N_T^k} P_e (w_{a(e)} - \bar{w}_{a(e)})^\top r_e \geq \mu, \quad k = 1, 2, \dots, K \quad (43)$$

which ensures the expected excess return must be at least μ under each of the k rival return scenarios, not just under one nominal forecast. The second is

$$P_e (w_{a(e)} - \bar{w}_{a(e)})^\top (\Lambda_i + r_e^\top r_e) (w_{a(e)} - \bar{w}_{a(e)}) \leq v_e, \quad e \in N_t, i = 1, \dots, I_e \quad (44)$$

which caps the risk contribution at each node under every rival covariance matrix to a threshold, v_e , ensuring risk doesn't blow up under any plausible risk model, not just the nominal one.

$$\begin{aligned} \min_w \quad & \gamma \sum_{t=1}^T \alpha_t \sum_{e \in N_t} v_e - \mu + \lambda \sum_{t=1}^T \sum_{e \in N_t} b_e s_e \\ \text{subject to} \quad & \mathbf{1}' b_0 - \mathbf{1}' s_0 = 1 - \mathbf{1}' p, \\ & p + (1 - c_b) b_0 - (1 + c_s) s_0 = w_0, \\ & r_e \circ w_{a(e)} + (1 - c_b) \circ b_e - (1 + c_s) \circ s_e = w_e, \quad e \in N_I, \\ & \mathbf{1}' b_e - \mathbf{1}' s_e = 0, \quad e \in N_I, \\ & \sum_{e \in N_T^k} P_e (w_{a(e)} - \bar{w}_{a(e)})' r_e \geq \mu, \quad e \in N_I, \quad k = 1, \dots, K, \\ & P_e (w_{a(e)} - \bar{w}_{a(e)})' (\Lambda_i + r_e' r_e) (w_{a(e)} - \bar{w}_{a(e)}) \leq v_e, \quad e \in N_I, \quad i = 1, \dots, I_e, \\ & w_e^L \leq w_e \leq w_e^U, \quad e \in N, \\ & 0 \leq b_e \leq b_e^U, \quad e \in N_I \cup \{0\}, \\ & 0 \leq s_e \leq s_e^U, \quad e \in N_I \cup \{0\}. \end{aligned} \quad (45)$$

Challenge of MSP problems: (1) sequential decision-making under uncertainty; (2) curse of dimensionality: exponential growth of scenarios and states; (3) wide range of solution approaches.

Scenario Tree-Based Deterministic Equivalent (DE) Method: In the past section, it introduced the Scenario Tree-Based Deterministic Equivalent (DE) Methods. In summary, this method discretizes the underlying (possibly continuous) stochastic process $\xi_1, \dots, \xi_T$ into a finite scenario tree and converts the resulting tree-based problem into a deterministic equivalent (DE), a single, large, ordinary optimization problem (LP or QP) with no explicit randomness left in it, since every scenario/node is now just a fixed, enumerated data point weighted by its probability.

Pros of the method:

- Conceptually simple and works with standard solvers. Once you have the DE formulation (a big LP or QP), you can hand it directly to off-the-shelf solvers (CPLEX, Gurobi, etc.) without needing specialized algorithms.
- Provides globally optimal solutions. But crucially, only for the approximated tree, not necessarily for the true underlying continuous stochastic process. This is an important caveat: solving the DE exactly gives you the exact optimum of your discretized approximation, but that approximation itself

introduces error relative to the true problem.

Cons:

- Size grows exponentially with stages.
- Quickly becomes intractable for large-scale problems. Practically speaking, once you have more than a handful of stages with realistic branching, the resulting DE becomes too large for even good solvers to handle in reasonable time.

Decomposition methods 1. Nested Benders Decomposition (Nested L-Shaped Method):

Standard Benders decomposition is a classical technique for two-stage stochastic/large LPs, where you split the problem into a master problem (first-stage decisions) and a subproblem (second-stage recourse), then iterate between them.

Nested means this idea is extended recursively to more than two stages. Decomposes by time stages and scenarios, that is, break the problem apart along both the time dimension (stage by stage) and the scenario dimension (branch by branch), and solve a master problem at each stage, which is linked to subproblems at the next stage, nested one inside another, mirroring the tree structure itself. The core mechanism of Benders-style decomposition is iteratively generate feasibility and optimality cuts. We solve a relaxed master problem, then use the subproblem's solution to generate a cut (a linear inequality) that either rules out an infeasible master solution (feasibility cut), or tightens the approximation of the true cost/value (optimality cut). It's a genuine improvement over solving the full DE directly, since we do not need to hold the entire tree in memory at once. But since it still requires enumerating/processing every scenario/node in the tree eventually, it inherits some of the same scalability limits when the tree itself is enormous.

2. Stochastic Dual Dynamic Programming (SDDP): it is also a sampling-based extension of nested Benders. Instead of working through the entire scenario tree (even via decomposition), SDDP only works with a manageable number of sampled paths through the tree.

- Forward pass is to simulate sample paths. Simulate a set of sample trajectories through the tree (from root to a leaf), using the current approximation of future costs, and record the decisions/states along each path.
- Backward pass is to generate cuts (value function approximations). Moving backward from the final stage to the first, use the sampled trajectories to build (or refine) an approximation of the cost-to-go function at each stage, essentially Benders-style cuts, but built incrementally from a sample of scenarios rather than the entire tree.
- **Pros:** handles large-scale, many-stage problems efficiently and avoids explicit enumeration of full scenario tree.
- **Cons:** typically assumes convexity (linear problems) and requires stagewise independence (in basic form), which means assuming that the randomness at each stage is independent of previous stages.

3. Hybrid / Advanced Decomposition: improve convergence and scalability.

- **Adaptive Benders + SDDP combinations:** mixing the cut-generation logic of Benders with the sampling efficiency of SDDP.
- **Multicut variants:** Instead of generating a single aggregate cut per iteration, generate multiple cuts (one per scenario/subproblem), and this can improve convergence speed at the cost of more computation per iteration.
- **Block-separable or multi-timescale problems:** adapting decomposition to problems with special structure, e.g., where some parts of the problem naturally separate into blocks, or where different parts of the problem evolve on different time scales (e.g., hourly operational decisions nested within monthly capacity decisions).

Stochastic Dynamic Programming (SDP): the main idea is from *Bellman equation*

$$V_t(s_t) = \min_{x_t} \{c_t + \mathbb{E}[V_{t+1}(s_{t+1})]\}, \quad (46)$$

where s_t is the state at time t , and captures all the information needed to make an optimal decision from this point forward; $V_t(s_t)$ is the value function, and the minimum expected total cost (or maximum expected value) achievable from state s_t onward, from time t to the end of the horizon; c_t is the cost incurred at the current stage from taking decision x_t ; and $\mathbb{E}[V_{t+1}(s_{t+1})]$ is the expected future cost-to-go, that is, how much expected cost to accumulate from $t + 1$ onward, given the state you transition into.

To know the best decision at time t , you need to know V_{t+1} (the value function one step ahead), and solve this backward in time, starting from the final stage T (where V_T is trivial/known) and working back to $t = 1$.

This method is based on the exact theoretical solution framework and is natural representation of sequential decisions but it suffers severely from curse of dimensionality.

Approximate Dynamic Programming (ADP): an extension of SDP. Since computing $V_t(s_t)$ exactly over the full state space is intractable for high-dimensional problems, ADP instead approximates the value function or the policy directly, using several possible techniques:

- **Value function approximation:** Instead of computing $V_t(s_t)$ exactly for every state, approximate it with some parametric or flexible functional form (e.g., linear, piecewise-linear, or neural-network-based approximation), which is conceptually similar to how SDDP builds up cut-based approximations of the cost-to-go function rather than computing it exactly everywhere.
- **Policy function approximation:** Instead of approximating the value function, directly approximate the decision rule/policy $x_t(s_t)$ itself with some functional form.
- **Lookahead policies and so on:** make decisions by only optimizing over a limited horizon into the future (rather than the full remaining horizon), often combined with an approximation of the tail value, a common practical simplification when the full backward recursion is infeasible.

This method scales to high-dimensional problems by avoiding exact enumeration of the state space, and ADP sidesteps the exponential blow-up that plagues exact SDP, and the method also links MSP with reinforcement learning / ML approaches, but the quality depends on approximation design.

Sampling and simulation-based methods: (1) SAA, shown in 2.3, replace expectation with sample average, and solves the resulting deterministic problem. (2) Monte Carlo Tree Methods use sampled scenarios dynamically and often combine with SDDP or ADP.

Monte Carlo Tree Methods: Rather than fixing a static scenario tree in advance, sample scenarios on the fly as the algorithm runs, which allows more flexibility, e.g., focusing sampling effort on promising or uncertain regions of the decision space, similar to Monte Carlo Tree Search (MCTS) in reinforcement learning/game-playing contexts. Besides, SDDP’s forward pass literally simulates sample paths through the tree, and ADP often relies on Monte Carlo sampling to estimate value functions when exact computation is infeasible. This shows these methods aren’t mutually exclusive, and sampling is a general tool that gets embedded within decomposition and DP-based algorithms, rather than a wholly separate, standalone technique.

Scenario reduction & tree approximation: it reduces problem size before solving, which is fundamentally a preprocessing step, not a solution algorithm itself.

Scenario reduction (forward/backward selection) starts with a large set of scenarios and systematically remove less important/representative ones (forward selection builds up a reduced set by adding the most important scenarios one at a time; backward selection starts with the full set and removes the least important ones), while adjusting probabilities of the remaining scenarios to preserve statistical fidelity as much as possible.

Clustering groups similar scenarios together and represents each cluster by a single representative scenario.

These are important preprocessing step for MSP tractability, and scenario reduction is meant to be used in combination with other solution methods (DE, decomposition, DP), making that downstream method

faster and more tractable.

Policy-based and decision rule approaches: The core concept here differs fundamentally from everything discussed so far. Rather than seeking the absolutely optimal decision for every possible state or scenario, the search space is restricted in advance to a class of decision rules that are computationally tractable. In the previous discussion, the true multi-stage decision x_t is technically an arbitrary function of the observed history $\xi_{[t-1]}$ and $x_t = x_t(\xi_{[t-1]})$, with no constraint on the functional form. This generality makes it hard to solve it exactly. Now, instead of allowing $x_t(\cdot)$ to be any function, restrict it to belonging to a specific, tractable parametric family, turning an infinite-dimensional search over arbitrary functions into a finite-dimensional search over just a few parameters. Three specific forms:

- **Linear decision rules:** force the decision to be a linear function of the observed uncertainty, $x_t(\xi_{[t-1]}) = A_t \xi_{[t-1]} + b_t$.
- **Affine/adaptive policies:** a slight generalization of linear rules, and **adaptive** emphasizes that the rule still adjusts based on observed information (preserving non-anticipativity), just through a restricted functional form rather than a fully general one.

The policy-based and decision rule approaches make the problem tractable (convex approximation) by restricting it to linear/affine forms; the resulting optimization problem over the parameters often becomes a convex optimization problem (e.g., an LP or SOCP), which can be solved efficiently with standard tools. Also they are suitable for high-dimensional uncertainty, just fitting a small number of coefficients. However, since x_t is assumed to be linear/affine, the resulting solution is generally not the true optimal policy, which sacrifices some of that potential performance (**suboptimal**).

Data-driven & ML methods:

- **Learning-Based Policies** generalize the policy function approximation idea flagged in ADP, but instead of restricting to linear/affine forms, it uses machine learning models as the approximation architecture, which is more flexible than a simple linear function.
 - **Supervised learning from scenario trees** uses an already-built scenario tree as training data and treats the pairs (state/history $\rightarrow$ optimal decision) computed on the tree as labeled examples, then trains a supervised learning model (e.g., a neural network) to predict the decision as a function of the state, generalizing beyond just the specific nodes in the tree. Once trained, the learned policy can be evaluated at any state, not just the discrete nodes originally in the tree and this addresses the tree’s inherent limitation of only covering a finite, pre-specified set of scenarios.
 - **Reinforcement learning (RL)** is explicitly the connection with the ADP discussion. RL agents learn an optimal policy through trial-and-error interaction with a (simulated) environment, using reward signals, without needing an explicit scenario tree or an explicit model of the transition dynamics at all. This is essentially ADP taken to its modern extreme: instead of hand-designing a value function approximation architecture, let a flexible RL algorithm (e.g., deep Q-learning, policy gradient methods) learn both the value function and the policy directly from simulated experience.
- **Kernel / Functional Methods** are more mathematically rigorous and function-space-based to the same underlying idea by representing flexible decision rules without restricting to a specific simple parametric form.
 - **Represent decisions in function spaces:** Rather than assuming decisions are literally linear or committing to a specific neural network architecture, this approach represents the decision rule $x_t(\cdot)$ as an element of a general (often infinite-dimensional) function space e.g., a reproducing kernel Hilbert space (RKHS), allowing much richer, nonlinear functional forms while still keeping the optimization problem mathematically tractable through kernel methods’ well-established theory.
 - **Use stochastic gradient optimization:** Since the function space may be very high- (or infinite-) dimensional, these methods are typically optimized using stochastic gradient descent (SGD) or similar first-order methods. The same optimization workhorse is used throughout modern machine learning, rather than classical LP/QP solvers used for the DE approach or the decomposition methods.

- **Achieve near-optimal solutions in data-rich settings:** The key promise here is that, given enough data (many samples/scenarios), these flexible functional methods can get very close to the true optimal policy, addressing precisely the ”suboptimal (restricted policy class)” weakness flagged for simple decision rules, at the cost of needing more data and more sophisticated (though still tractable) optimization machinery.

Table 1: Comparison of Solution Methods for Multi-Stage Stochastic Programming

Approach	Strength	Weakness	Typical Use
Deterministic equivalent	Exact for given scenarios	Explodes in size	Small/medium problems
Nested Benders Decomposition	Structured decomposition	Still tree-dependent	Medium MSP
SDDP	Scalable, industry standard	Convexity assumptions	Large-scale (energy, finance)
SDP	Exact theory	Intractable in practice	Small state space
ADP	Scalable, flexible	Approximation error	High-dimensional problems
Decision rules	Tractable	Suboptimal policies	Fast approximations
ML/data-driven	Flexible, efficient	Requires training data	Large-scale MSP

Comparison of Solution Methods Remarks

- **Decomposition methods, especially SDDP**, dominate practical large-scale MSP, thanks to their scalability;
- **approximation methods (ADP, decision rules, ML)** are increasingly essential when state/action spaces grow too large for exact or decomposition-based approaches;
- **scenario trees and scenario reduction** remain the foundational representation underlying most of these methods;
- **Machine Learning** represents the emerging frontier, offering flexibility and data-efficiency at the cost of requiring training data;
- across all approaches, the central and unresolved challenge is balancing **accuracy vs. computational tractability** in high-dimensional, uncertain, multi-stage settings.

Literature: [Birge and Louveaux \(1997\)](#); [Shapiro et al. \(2014\)](#); [Pflug and Pichler \(2014\)](#).

3. Robust Optimization

Robust optimization is to find a feasible solution x that achieve the best worst-case objective. Drop distributional assumptions entirely; let r merely belong to an *uncertainty set* $\mathcal{U}$. For example,

$$\max_{0 \leq x \leq 1000} \min_{r \in \mathcal{U}} (r - 0.01)x = (r_{\min} - 0.01)x. \quad (47)$$

The worst-case guarantees matter in applications like disaster management and cancer therapy planning, and conservativeness is controlled through the choice of $\mathcal{U}$.

3.1 Uncertainty sets

(for $\mathbf{r} = (r_1, \dots, r_n)$): there is no distribution information and no expected objective.

- Scenario set from historical data:

$$\mathcal{U}_1 = \{r_s\}_{s=1, \dots, S} \quad (48)$$

- Box: from statistical information (mean, variance, etc.) and (or) expert knowledge,

$$\mathcal{U}_2 = \{\mathbf{r} : r_i \in [\bar{r}_i - \Delta r_i, \bar{r}_i + \Delta r_i]\} \quad (49)$$

where Δr_i is flexible, e.g., based on standard deviation.

- Norm-based:

$$\mathcal{U}_{\ell_p} = \{\mathbf{r} : \|\mathbf{r} - \bar{\mathbf{r}}\|_p \leq \Delta\} \quad (50)$$

$$\ell_1 \text{ gives a polyhedral (diamond) set, } \ell_2 \text{ an ellipsoidal set.} \quad (51)$$

- Budget uncertainty sets: at most $\lfloor \Gamma \rfloor$ parameters deviate fully by Δr_i , one more by $(\Gamma - \lfloor \Gamma \rfloor)\Delta r_i$, with budget $\Gamma \in [0, n]$. The Bertsimas-Sim construction is given as an exercise.

3.2 Robust constraints

Robust constraints should follow the **immunization** principle. A constraint with uncertain data must hold for every realization

$$\sum_{i=1}^n r_i x_i \geq r_t, \forall \mathbf{r} \in \mathcal{U} \iff \min_{\mathbf{r} \in \mathcal{U}} \sum_{i=1}^n r_i x_i \geq r_t \quad (52)$$

So both robust objectives and robust constraints reduce to an inner optimization problem $f(\mathbf{x}) = \min_{\mathbf{r} \in \mathcal{U}} \sum_i r_i x_i$.

Robust objective and constraint

$$\begin{aligned} \max_{\mathbf{x} \in \mathcal{X}} \min_{\mathbf{r} \in \mathcal{U}} \sum_i r_i x_i, \\ \min_{\mathbf{r} \in \mathcal{U}} \sum_i r_i x_i \geq r_t. \end{aligned} \quad (53)$$

For some $\mathcal{U}$, $f(\mathbf{x})$ can be computed analytically. When $f(\mathbf{x})$ can't be computed analytically but $\mathcal{U}$ is polyhedral, the inner min is an LP. Strong duality ($Z_P = Z_D$ when both feasible) lets us replace the inner minimization by its dual maximization:

$$\max_{\mathbf{x} \in \mathcal{X}} \min_{\mathbf{r} \in \mathcal{U}} \sum_i r_i x_i = \max_{\mathbf{x} \in \mathcal{X}, \mathbf{p} \in \mathcal{P}(\mathbf{x})} g(\mathbf{p}). \quad (54)$$

The min-max becomes a single max over $(\mathbf{x}, \mathbf{p})$, a tractable problem with auxiliary dual variables and constraints.

3.3 Duality

To compute $f(\mathbf{x})$, it needs the strong duality and the dual problem of the inner optimization problem.

$$\begin{array}{ll|ll}
(P): & Z_P = \max c^T x & & (D): & Z_D = \min b^T p \\
& \text{s.t. } Ax \leq b & & & \text{s.t. } p \geq 0 \\
& x \geq 0 & & & A^T p \geq c.
\end{array} \tag{55}$$

Theorem 3.1. Strong Linear Duality

Exactly one of the following holds for the primal-dual problem pair (P) and (D):

- Both problems are feasible and optimal solutions $\mathbf{x}^*$ and $\mathbf{p}^*$ are achievable with $Z_P = Z_D$.
- One problem is unbounded, and the other is infeasible.
- Both problems are infeasible.

To get the dual problem, we need to build the Lagrangian and choose the sign of each multiplier so that weak duality (the bounding inequality) holds.

Consider a primal problem in maximization form, with constraints of the three types. Let the row indices be split into the sets $I_{\leq}$, $I_{=}$, and $I_{\geq}$:

$$Z_P = \max_{\mathbf{x}} c^T \mathbf{x} \quad \text{s.t.} \quad \mathbf{a}_i^T \mathbf{x} \in \{\leq, =, \geq\} b_i. \tag{56}$$

We want to construct, for each choice of multipliers $\mathbf{p}$, an upper bound on Z_P . Since the primal is a maximization problem, the dual must bound it from above. The bound comes from adding p_i times constraint i to the objective.

For any feasible $\mathbf{x}$, we want

$$c^T \mathbf{x} \leq c^T \mathbf{x} + \sum_i p_i (b_i - \mathbf{a}_i^T \mathbf{x}). \tag{57}$$

Equivalently, every added term

$$p_i (b_i - \mathbf{a}_i^T \mathbf{x}) \tag{58}$$

must be nonnegative.

We check each constraint type:

$$\text{If } \mathbf{a}_i^T \mathbf{x} \leq b_i \Rightarrow b_i - \mathbf{a}_i^T \mathbf{x} \geq 0 \Rightarrow p_i \geq 0. \tag{59}$$

$$\text{If } \mathbf{a}_i^T \mathbf{x} = b_i \Rightarrow b_i - \mathbf{a}_i^T \mathbf{x} = 0 \Rightarrow p_i \text{ is free.} \tag{60}$$

$$\text{If } \mathbf{a}_i^T \mathbf{x} \geq b_i, \Rightarrow b_i - \mathbf{a}_i^T \mathbf{x} \leq 0, \Rightarrow p_i \leq 0. \tag{61}$$

Rearranging the upper bound gives

$$c^T \mathbf{x} \leq b^T \mathbf{p} + (\mathbf{c} - \mathbf{A}^T \mathbf{p})^T \mathbf{x}. \tag{62}$$

The bound is useful only if the term

$$(\mathbf{c} - \mathbf{A}^T \mathbf{p})^T \mathbf{x} = \sum_j (c_j - \mathbf{A}_j^T \mathbf{p}) x_j \tag{63}$$

cannot blow up.

Equivalently, in the Lagrangian dual function

$$g(\mathbf{p}) = \mathbf{b}^T \mathbf{p} + \max_{x_j \in \text{sign domain}} \sum_j (c_j - \mathbf{A}_j^T \mathbf{p}) x_j, \quad (64)$$

the inner maximization over each x_j must be finite. Define

$$d_j := c_j - \mathbf{A}_j^T \mathbf{p}.$$

We check each variable type.

$$\text{If } x_j \geq 0, \Rightarrow \max_{x_j \geq 0} d_j x_j = \begin{cases} 0, & d_j \leq 0, \\ +\infty, & d_j > 0. \end{cases} \Rightarrow d_j \leq 0, \Rightarrow \mathbf{A}_j^T \mathbf{p} \geq c_j. \quad (65)$$

Hence, the dual constraint is in $\geq$ form.

$$\text{If } x_j \text{ is free} \Rightarrow x_j \in \mathbb{R}, \Rightarrow \max_{x_j \in \mathbb{R}} d_j x_j \text{ is finite only if } d_j = 0 \Rightarrow \mathbf{A}_j^T \mathbf{p} = c_j. \quad (66)$$

Hence the dual constraint is an equality.

$$\text{If } x_j \leq 0, \text{ then the inner maximization is finite only if } d_j \geq 0. \Rightarrow \mathbf{A}_j^T \mathbf{p} \leq c_j. \quad (67)$$

Hence the dual constraint is in $\leq$ form.

Therefore the dual problem is

$$\min_{\mathbf{p}} \mathbf{b}^T \mathbf{p} \quad (68)$$

subject to the corresponding sign restrictions on $\mathbf{p}$ and the dual feasibility constraints. We minimize because we want the tightest upper bound.

A natural $\leq$ constraint gives a naturally signed multiplier:

$$p_i \geq 0. \quad (69)$$

A nonnegative primal variable

$$x_j \geq 0 \Rightarrow \mathbf{A}_j^T \mathbf{p} \geq c_j. \quad (70)$$

Equality constraints correspond to free multipliers:

$$\mathbf{a}_i^T \mathbf{x} = b_i \iff p_i \text{ is free.} \quad (71)$$

Free variables correspond to equality constraints:

$$x_j \text{ is free} \iff \mathbf{A}_j^T \mathbf{p} = c_j. \quad (72)$$

A wrong-direction constraint,

$$\mathbf{a}_i^T \mathbf{x} \geq b_i, \text{ forces } p_i \leq 0. \quad (73)$$

A wrong-signed variable,

$$x_j \leq 0, \text{ forces } \mathbf{A}_j^T \mathbf{p} \leq c_j. \quad (74)$$

For a minimization primal, everything flips. The natural constraint is now $\geq$, giving $p_i \geq 0$.

Duality in Matrix Form

$$\begin{array}{c|c}
(P): \quad Z_P = \max c^T x & (D): \quad Z_D = \min b^T p \\
\text{s.t.} \quad Ax \leq b & \text{s.t.} \quad p \geq 0 \\
\quad \quad \quad x \geq 0 & \quad \quad \quad A^T p \geq c \\
\hline
\sum_{j=1}^n a_{ij} x_j - b_i \leq 0 & p_i \geq 0 \\
\hline
x_j \geq 0 & \sum_{i=1}^m p_i a_{ij} - c_j \geq 0
\end{array} \tag{75}$$

Example: the inner problem, x fixed, r_1 , and r_2 are the decision variables

$$\begin{array}{ll}
\min_{r_1, r_2} & x_1 r_1 + x_2 r_2 \\
\text{s.t.} & \begin{cases} r_1 + r_2 \leq \Delta + \bar{r}_1 + \bar{r}_2, & (p_1), \\ r_1 - r_2 \leq \Delta + \bar{r}_1 - \bar{r}_2, & (p_2), \\ -r_1 + r_2 \leq \Delta - \bar{r}_1 + \bar{r}_2, & (p_3), \\ -r_1 - r_2 \leq \Delta - \bar{r}_1 - \bar{r}_2, & (p_4). \end{cases}
\end{array} \tag{76}$$

The four constraints are the unfolding of

$$|r_1 - \bar{r}_1| + |r_2 - \bar{r}_2| \leq \Delta. \tag{77}$$

Min primal with $\leq$ constraints implies $p_i \leq 0$

For a minimization problem, we need a lower bound, i.e. adding $p_i (b_i - a_i^T r)$ must be ≤ 0 . Since $b_i - a_i^T r \geq 0$ under feasibility, we need $p_i \leq 0$.

The variables r_1, r_2 are free, therefore they give equality constraints in the dual, one per primal variable. The coefficient of each dual constraint is read from the corresponding column of the constraint matrix, and the right-hand side is the primal objective coefficient.

For the dual objective, change min to max, and the objective coefficients are the primal RHS values b_i :

$$g(p) = (\bar{r}_1 + \bar{r}_2 + \Delta)p_1 + (\Delta + \bar{r}_1 - \bar{r}_2)p_2 + (\Delta - \bar{r}_1 + \bar{r}_2)p_3 + (\Delta - \bar{r}_1 - \bar{r}_2)p_4. \tag{78}$$

Then the dual problem is

$$\begin{array}{ll}
\max_{p_1, p_2, p_3, p_4} & (\bar{r}_1 + \bar{r}_2 + \Delta)p_1 + (\Delta + \bar{r}_1 - \bar{r}_2)p_2 + (\Delta - \bar{r}_1 + \bar{r}_2)p_3 + (\Delta - \bar{r}_1 - \bar{r}_2)p_4 \\
\text{s.t.} & p_1 + p_2 - p_3 - p_4 = x_1, \\
& p_1 - p_2 + p_3 - p_4 = x_2, \\
& p_1, p_2, p_3, p_4 \leq 0.
\end{array} \tag{79}$$

For the robust optimization

$$\max_{x \in \mathcal{X}} \min_{r \in \mathcal{U}_{\ell_1}} r_1 x_1 + r_2 x_2 = \max_{x \in \mathcal{X}} \max_{p \in \mathcal{P}(x)} g(p), \tag{80}$$

equivalently

$$\begin{aligned}
& \max_{x,p} g(p) \\
& \text{s.t. } p \in \mathcal{P}(x), \\
& x \in \mathcal{X}.
\end{aligned} \tag{81}$$

where

$$\begin{aligned}
g(p) &= (\bar{r}_1 + \bar{r}_2 + \Delta)p_1 + (\Delta + \bar{r}_1 - \bar{r}_2)p_2 + (\Delta - \bar{r}_1 + \bar{r}_2)p_3 + (\Delta - \bar{r}_1 - \bar{r}_2)p_4, \\
\mathcal{P}(x) &= \{(p_1, p_2, p_3, p_4) \leq 0 : p_1 + p_2 - p_3 - p_4 = x_1, p_1 - p_2 + p_3 - p_4 = x_2\}.
\end{aligned} \tag{82}$$

To reformulate a robust constraint rather than a robust objective, there is a chain of three equivalences: (1) swap min for max by strong duality, (2) replace max by $\exists$ and (3) absorb $\exists$ into the feasible set (an existential constraint is implemented by just promoting p to a decision variable).

$$\begin{aligned}
\min_{r \in \mathcal{U}} r_1 x_1 + r_2 x_2 \geq r_t &\iff \max_{p \in \mathcal{P}(x)} g(p) \geq r_t \\
&\iff \exists p \in \mathcal{P}(x) : g(p) \geq r_t \\
&\iff \begin{cases} g(p) \geq r_t, \\ p \in \mathcal{P}(x). \end{cases}
\end{aligned} \tag{83}$$

3.4 Distributionally Robust Optimization (DRO)

Distributionally robust optimization keeps the stochastic model's expectation objective but applies the robust model's worst-case principle to the distribution itself. In DRO, the distribution P is only known to lie in an ambiguity set $\mathcal{P}$,

$$\max_{x \in \mathcal{X}} \min_{P \in \mathcal{P}} \mathbb{E}_P[f(x; \xi)] \tag{84}$$

Ambiguity Sets

- **Moment-based ambiguity sets:** based on the summary statistics but not the full shape

$$\mathcal{P}_{\gamma_1, \gamma_2}(\mu, \Gamma) = \left\{ P : (\mathbb{E}_P[\xi] - \mu)^T \Gamma^{-1} (\mathbb{E}_P[\xi] - \mu) \leq \gamma_1, \quad \text{Cov}_P[\xi] \preceq \gamma_2 \Gamma \right\}. \tag{85}$$

The first condition restricts the mean to an ellipsoid centered at μ , while the second condition constrains the covariance in the sense of the positive semi-definite (Loewner) order—specifically, $\gamma_2 \Gamma - \text{Cov}_P \succeq 0$. When $\gamma_1 = 0$ and $\gamma_2 = 1$, the mean is fixed at μ and the covariance is bounded by Γ (i.e., the Delage–Ye set).

- **Marginal-based ambiguity sets:** based on the information of marginal distributions.

$$\mathcal{P}(P_1, \dots, P_m) = \{P \mid \Pi_{M_i}(P) = P_i, \ M_i \subset \{1, \dots, n\}\}. \tag{86}$$

This is a Fréchet class and the ambiguity is purely about the copula. For you this should ring loud bells: this is exactly the setting of copula bounds, comonotonicity, and dependence uncertainty in risk aggregation. The worst-case joint distribution for many objectives is a comonotone or antimonotone coupling, and the dual of optimizing over this set is a Monge–Kantorovich-type problem.

- **Divergent-based ambiguity sets:** A ball around a reference distribution P_0 (usually the empirical measure $\hat{P}_N$),

$$\mathcal{P}_\epsilon(P_0) = \{P \mid d(P, P_0) \leq \epsilon\}. \tag{87}$$

There are two main choices of the distance

- φ -divergences (KL, χ^2 , etc.): $d(P, P_0) = \int \varphi(dP/dP_0) dP_0$. Cheap reformulations, but there is a limitation, that is, P must be absolutely continuous w.r.t. P_0 . If P_0 is empirical, every $P \in \mathcal{P}$ lives on the observed data points only; the adversary can reweight history but never imagine an unseen scenario.
- Wasserstein distance: $d_{W,p}(P_1, P_2) = \inf_{\pi \in \mathcal{P}} (\mathbb{E}_\pi \|\xi_1 - \xi_2\|^p)^{1/p}$, ($p \geq 1$), the optimal transport cost required to transform one distribution into another. The ball contains distributions with different supports, and the adversary can perturb data points off their observed locations. Statistically,

measure-concentration results give a rate for choosing $\epsilon = \epsilon(N)$ so that the true P lies in the ball with high confidence, making the DRO value a legitimate out-of-sample performance guarantee.

How to reformulate? The reformulation logic is identical to the RO to dualize the inner adversarial problem, except the inner problem is now an infinite-dimensional LP over measures. Note that $\mapsto \mathbb{E}_P[f(\mathbf{x}; \boldsymbol{\xi})]$ it is linear in P (expectation is an integration), and the ambiguity set is defined by finitely many linear in P constraints (moments) plus normalization. So the inner problem

$$\min_P \int f(x; \xi) dP \quad \text{s.t.} \quad \int dP = 1, \quad \int \xi dP = \mu, \quad \int \xi \xi^T dP \preceq \Sigma, \quad P \succeq 0, \quad (88)$$

is an LP whose variables are the measure's values. Attach multipliers to each constraint (λ for normalization, q for the mean, $Q \succeq 0$ for the covariance) and the dual becomes choosing the multipliers to

$$\max_{\lambda, q, Q \succeq 0} \quad \lambda + q^T \mu + \langle Q, \Sigma \rangle \quad \text{s.t.} \quad \lambda + q^T \xi + \xi^T Q \xi \leq f(x; \xi), \quad \forall \xi. \quad (89)$$

The dual constraint says: find a quadratic function that minorizes $f(x; \cdot)$ pointwise; the dual objective is that the quadratic's expectation under the moment data. Weak duality is immediate (integrate the pointwise inequality against any feasible P); strong duality holds under a Slater condition (the moment vector is interior to the feasible moment cone). The $\forall \xi$ constraint is again a semi-infinite constraint, but for f piecewise linear in ξ it converts to LMIs via the S-lemma $\rightarrow$ an SDP. Same $\forall \rightarrow \exists$ collapse as the robust-constraint slide, just with a continuum of primal variables instead of finitely many.

For Wasserstein balls the same program yields the celebrated *Mohajerin Esfahani–Kuhn/Blanchet–Murthy duality*:

$$\min_{P: d_W(P, \hat{P}_N) \leq \epsilon} \mathbb{E}_P[f(x; \xi)] = \max_{\lambda \geq 0} \left\{ -\lambda \epsilon + \frac{1}{N} \sum_{i=1}^N \inf_{\xi} \left[f(x; \xi) + \lambda \|\xi - \hat{\xi}_i\| \right] \right\}, \quad (90)$$

a single scalar multiplier λ prices transport budget, and the inner inf is a *Moreau-type regularization* of f around each data point. For piecewise linear f this is again a finite LP/conic program. For many loss functions, Wasserstein-DRO is exactly equivalent to norm-regularized empirical optimization (e.g., regularized regression/SVM), that is, **robustness is regularization**, with ϵ playing the role of the penalty weight.

Example: consider the distributionally robust optimization and reformulate the robust counterpart,

$$\begin{aligned} & \max_{x \in \mathcal{X}} \min_{P \in \mathcal{P}} \mathbb{E}_P[(r - 0.01)x + 10] \\ & \text{s.t.} \quad 0 \leq x \leq 1000 \end{aligned} \quad (91)$$

where $\mathcal{P} = \mathcal{P}_{\gamma_1, \gamma_2}(\mu, \sigma^2)$.

In one dimension, the ambiguity set is

$$\mathcal{P} = \left\{ P : \frac{(\mathbb{E}_P[r] - \mu)^2}{\sigma^2} \leq \gamma_1, \quad \text{Var}_P(r) \leq \gamma_2 \sigma^2 \right\}. \quad (92)$$

$$\min \mathbb{E}_P[(r - 0.01)x + 10] = \left(\min_{P \in \mathcal{P}} \mathbb{E}_P[r] \right) x - 0.01x + 10, \quad x \geq 0. \quad (93)$$

The mean constraint says

$$\mathbb{E}_P[r] \in [\mu - \sqrt{\gamma_1} \sigma, \mu + \sqrt{\gamma_1} \sigma]. \quad (94)$$

So worst-case mean is

$$\mathbb{E}_P[r]^* = \mu - \sqrt{\gamma_1} \sigma. \quad (95)$$

The variance budget γ_2 plays no role whatsoever, because the objective is linear in r .

Substituting back, the DRO problem is now a deterministic LP:

$$\max_{0 \leq x \leq 1000} (\mu - \sqrt{\gamma_1} \sigma - 0.01) x + 10. \quad (96)$$

Linear in x , exactly as in the deterministic model, but with the ambiguity-adjusted return $r_{\min} := \mu - \sqrt{\gamma_1} \sigma$,

in place of r :

$$x^* = \begin{cases} 1000, & \mu - \sqrt{\gamma_1} \sigma > 0.01, \\ 0, & \mu - \sqrt{\gamma_1} \sigma < 0.01, \\ \text{anything in } [0, 1000], & \mu - \sqrt{\gamma_1} \sigma = 0.01. \end{cases} \quad (97)$$

The optimal value is

$$\max \{1000 (\mu - \sqrt{\gamma_1} \sigma - 0.01), 0\} + 10. \quad (98)$$

An alternative way is as following. Write the inner problem over measures:

$$\min_P \int (r - 0.01) x dP \quad \text{s.t.} \quad \int dP = 1 \ (\lambda), \quad \left| \int r dP - \mu \right| \leq \sqrt{\gamma_1} \sigma \ (q). \quad (99)$$

Dropping the inactive variance constraint, we split the two-sided mean constraint into

$$\mu - \sqrt{\gamma_1} \sigma \leq \int r dP \leq \mu + \sqrt{\gamma_1} \sigma. \quad (100)$$

Define the Lagrangian:

$$\begin{aligned} L(P; \lambda, q_-, q_+) &= \int (r - 0.01) x dP - \lambda \left(\int dP - 1 \right) \\ &\quad - q_- \left(\int r dP - (\mu - \delta) \right) - q_+ \left(\int r dP - (\mu + \delta) \right). \end{aligned} \quad (101)$$

For every primal-feasible P , we need $L \leq$ primal objective. That is, each subtracted term must be nonnegative. More precisely, we need the subtracted penalty to be nonpositive, so that

$$L(P) \leq \int (r - 0.01) x dP. \quad (102)$$

Check term by term for feasible P :

• λ -term:

$$\int dP - 1 = 0 \Rightarrow \lambda \text{ is free} \quad (103)$$

• q_- -term:

$$\int r dP - (\mu - \delta) \geq 0 \Rightarrow q_- \geq 0. \quad (104)$$

• q_+ -term:

$$\int r dP - (\mu + \delta) \leq 0, \text{ since } q_+(\cdot) \geq 0 \Rightarrow q_+ \leq 0. \quad (105)$$

Thus the lower-bound-generating multipliers satisfy $q_- \geq 0 \geq q_+$, λ is free.

Group L by what multiplies P :

$$L = \int \underbrace{[(r - 0.01)x - \lambda - (q_- + q_+)r]}_{=:h(r)} dP + \lambda + q_-(\mu - \delta) + q_+(\mu + \delta). \quad (106)$$

Now take $\inf_{P \geq 0} L$. Here P ranges over all nonnegative measures. The normalization constraint has already been dualized away, so the total mass is unconstrained.

- If $h(r_0) < 0$ for some point r_0 , put mass $t \rightarrow \infty$ at r_0 , and then $L \rightarrow -\infty$. This gives a useless bound.
- If $h(r) \geq 0$ for all r , the best choice is $P = 0$, or any measure supported where $h = 0$, giving $\int h dP = 0$.

So the dual function is finite if and only if $h(r) \geq 0$ pointwise, and then it equals the constant part. Writing

$$q := q_- + q_+ \quad (107)$$

, note that q is free in sign since it is the sum of a nonnegative and a nonpositive number. Therefore,

$$\begin{aligned} g(\lambda, q_-, q_+) &= \lambda + q_-(\mu - \delta) + q_+(\mu + \delta) \\ \text{s.t. } \lambda + qr &\leq (r - 0.01)x, \quad \forall r \in \mathbb{R}. \end{aligned} \quad (108)$$

Rewrite the constraint

$$(q - x)r + (\lambda + 0.01)x \leq 0, \quad \forall r \in \mathbb{R}. \quad (109)$$

An affine function of r is ≤ 0 on the entire real line iff its slope is zero and its intercept nonpositive:

$$q = x, \quad \lambda \leq -0.01x. \quad (110)$$

The dual problem is now:

$$\max_{\lambda, q_- \geq 0, q_+ \leq 0} \lambda + q_-(\mu - \delta) + q_+(\mu + \delta) \quad \text{s.t. } q_- + q_+ = x, \quad \lambda \leq -0.01x. \quad (111)$$

The objective increases in λ , so take it to its cap:

$$\lambda^* = -0.01x. \quad (112)$$

Maximize

$$q_-(\mu - \delta) + q_+(\mu + \delta) \quad \text{s.t. } q_- + q_+ = x, \quad q_- \geq 0 \geq q_+. \quad (113)$$

Since

$$q_+ = x - q_- \Rightarrow q_- \geq \max(x, 0) = x, \quad (114)$$

and $x \geq 0$,

$$q_-(\mu - \delta) + (x - q_-)(\mu + \delta) = x(\mu + \delta) - 2\delta q_-. \quad (115)$$

This expression is decreasing in q_- since $\delta > 0$. Hence we push q_- to its smallest feasible value:

$$q_-^* = x \Rightarrow q_+^* = 0. \quad (116)$$

Economically, this is natural because a long position fears low returns. Adding back the constant +10, the dual value is

$$-0.01x + x\mu - x\sqrt{\gamma_1}\sigma = (\mu - \sqrt{\gamma_1}\sigma - 0.01)x + 10, \quad (117)$$

and

$$\max_{0 \leq x \leq 1000} (\mu - \sqrt{\gamma_1}\sigma - 0.01)x + 10, \quad (118)$$

$$x^* = \begin{cases} 1000, & \mu - \sqrt{\gamma_1}\sigma > 0.01, \\ 0, & \mu - \sqrt{\gamma_1}\sigma < 0.01, \\ \text{anything in } [0, 1000], & \mu - \sqrt{\gamma_1}\sigma = 0.01. \end{cases} \quad (119)$$

3.5 Extension: connection with decision theory

Connection with Maxmin expected utility The DRO criterion $\min_{P \in \mathcal{P}} \mathbb{E}_P[\cdot]$ is Gilboa–Schmeidler maxmin expected utility: MEU says an ambiguity-averse agent evaluates acts by $\min_{P \in \mathcal{C}} \mathbb{E}_P[u]$ over a set of priors $\mathcal{C}$. DRO is MEU where the agent is the optimizer and $\mathcal{C}$ is a statistically calibrated ambiguity set. The two bodies of literature proved the same representation theorems from opposite directions—decision theory axiomatized the preferences, and optimization built the algorithms.

Connection with neo-additive Choquet expectations. A neo-additive Choquet expectation takes the form

$$\alpha \delta \sup f + (1 - \alpha) \delta \inf f + (1 - \delta) \mathbb{E}_{P_0}[f], \quad (120)$$

where P_0 is a reference probability measure, $\delta \in [0, 1]$ captures the degree of ambiguity, and $\alpha \in [0, 1]$ governs optimism versus pessimism. This representation admits a natural robust interpretation. Consider the contamination class

$$\mathcal{P}_\delta(P_0) = \{(1 - \delta)P_0 + \delta Q : Q \in \Delta(\Omega)\}, \quad (121)$$

where $\Delta(\Omega)$ denotes the set of all probability measures on the state space. Then, for any bounded payoff f ,

$$\min_{P \in \mathcal{P}_\delta(P_0)} \mathbb{E}_P[f] = (1 - \delta) \mathbb{E}_{P_0}[f] + \delta \inf f, \quad (122)$$

since the contaminating distribution Q can assign all its mass to states where f attains its lowest value. Similarly,

$$\max_{P \in \mathcal{P}_\delta(P_0)} \mathbb{E}_P[f] = (1 - \delta) \mathbb{E}_{P_0}[f] + \delta \sup f. \quad (123)$$

Thus, the neo-additive Choquet expectation can be viewed as a convex combination of the optimistic and pessimistic robust evaluations over the same contamination neighborhood. In this sense, the neo-additive model is closely related to the classical ϵ -contamination neighborhoods used in robust statistics, going back to Huber. It can also be interpreted as a total-variation-flavored ambiguity set around the benchmark model P_0 , since a fraction δ of the probability mass is allowed to be arbitrarily misspecified.

Dynamic Inconsistency A key difficulty arises when extending static DRO or maxmin preferences to a dynamic model. A static robust criterion has the form

$$\inf_{P \in \mathcal{P}} \mathbb{E}_P[X], \quad (124)$$

where X is a terminal payoff. In general, the worst-case measure chosen at time t need not be consistent with the worst-case measure chosen at time $t + 1$. That is, the continuation plan that is optimal from the perspective of time t may no longer be optimal after new information arrives. This leads to dynamic inconsistency.

Formally, the problem is that one cannot generally write

$$\inf_{P \in \mathcal{P}} \mathbb{E}_P[X] \quad (125)$$

as a recursively consistent dynamic evaluation unless the ambiguity set satisfies additional stability properties. In particular, the conditional ambiguity sets must be compatible over time. Otherwise, the dynamic programming principle may fail.

The standard condition used to restore dynamic consistency is rectangularity. Rectangularity requires that the set of priors can be decomposed into conditional one-period ambiguity sets that can be recombined freely across histories. Under rectangularity, the robust evaluation admits a recursive representation of the form

$$V_t = \inf_{P_t \in \mathcal{P}_t} \mathbb{E}_{P_t}[V_{t+1} \mid \mathcal{F}_t]. \quad (126)$$

This recursive structure is essential for dynamic programming, HJB equations, and impulse-control formulations. The ambiguity set must be chosen so that it is dynamically consistent; a policy selected at

time t may be abandoned at time $t+1$ after the ambiguity set is recomputed. Without rectangularity or an equivalent consistency condition, the dynamic control problem may not be well-defined in the usual recursive sense.

3.6 Application: robust multi-period production planning under budgeted demand uncertainty

A firm plans production of a product over periods $t = 1, \dots, T$. Each period it chooses how much to produce, $x_t \geq 0$, at unit cost P . Demand D_t arrives each period. If cumulative production runs ahead of cumulative demand, the surplus is stored at holding cost H per unit per period; if it runs behind, the shortfall is backlogged at penalty B per unit per period. The goal is to minimize total cost.

The commercial setting is a new product launch. This means that the crucial input, demand, is not known, and there is no history from which to estimate a distribution. At best, the firm has a forecast $\bar{D}_t$ and a plausible deviation $\hat{d}_t$ for each period.

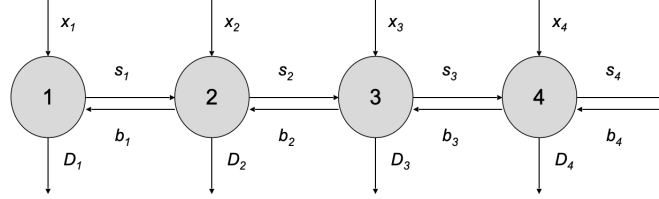


Figure 2: visualization

If we assume D_t is known at first, the natural first model tracks stock s_t and backlog b_t through a flow-balance equation:

$$\begin{aligned} \min \quad & \sum_{t=1}^T (Px_t + Hs_t + Bb_t) \\ \text{s.t.} \quad & x_t + s_{t-1} - b_{t-1} - s_t + b_t = D_t, \quad \forall t, \\ & x_t, s_t, b_t \geq 0, \\ & s_0 = b_0 = 0. \end{aligned} \tag{127}$$

Once demand is uncertain, demand appears in equality constraints.

With setup costs, one may add binary variables $y_t \in \{0, 1\}$ and impose

$$0 \leq x_t \leq My_t. \tag{128}$$

With capacity constraints, one may write

$$x_t \leq C_t y_t. \tag{129}$$

With $H, B > 0$, it is never optimal to hold stock and backlog simultaneously. Hence, at optimum,

$$s_t b_t = 0. \tag{130}$$

Therefore, the pair (s_t, b_t) can be collapsed into the net inventory

$$I_t = s_t - b_t. \tag{131}$$

Then

$$s_t = \max\{I_t, 0\}, \quad b_t = \max\{-I_t, 0\}. \tag{132}$$

The balance equation becomes

$$x_t + I_{t-1} - I_t = D_t. \tag{133}$$

The period inventory cost becomes

$$\max\{HI_t, -BI_t\}. \tag{134}$$

This convex piecewise-linear cost can be linearized with an epigraph variable c_t :

$$c_t \geq HI_t, \quad c_t \geq -BI_t. \quad (135)$$

Then, eliminate the state variable. From

$$I_t = x_t + I_{t-1} - D_t, \quad I_0 = 0, \quad (136)$$

we obtain

$$I_t = \sum_{j=1}^t (x_j - D_j). \quad (137)$$

Substituting this expression into the epigraph formulation gives

$$\begin{aligned} \min \quad & \sum_{t=1}^T (Px_t + c_t) \\ \text{s.t.} \quad & c_t \geq H \sum_{j \leq t} (x_j - D_j), \quad \forall t, \\ & c_t \geq B \sum_{j \leq t} (D_j - x_j), \quad \forall t, \\ & x_t \geq 0. \end{aligned} \quad (138)$$

In the deterministic world, these formulations are equivalent. However, under uncertainty, they are no longer equally useful. In the formula (127), demand appears in equality constraints. A static robust solution cannot satisfy

$$x_t + s_{t-1} - b_{t-1} - s_t + b_t = D_t \quad (139)$$

for all D_t in a non-singleton uncertainty set, unless some variables are made adjustable functions of demand. By contrast, in formulation (138), demand appears only in inequality constraints through cumulative sums. Inventory has been eliminated as a state variable and is now represented by the identity

$$I_t = \sum_{j \leq t} (x_j - D_j). \quad (140)$$

This makes static robustification possible.

With no distribution available, knowledge about demand is encoded as a set. Each period's demand is a nominal value plus a bounded perturbation:

$$D_t = \bar{D}_t + \hat{d}_t z_t, \quad -1 \leq z_t \leq 1. \quad (141)$$

A box uncertainty set protects against the extreme case where every period simultaneously deviates to its worst value. To reduce this conservatism, the Bertsimas-Sim budget of uncertainty limits the total deviation that the adversary may spend.

Because the constraints in (138) involve cumulative demand, the budget is also imposed cumulatively:

$$U = \left\{ D : D_t = \bar{D}_t + \hat{d}_t z_t, |z_t| \leq 1, \sum_{j=1}^t |z_j| \leq \Gamma_t, \forall t \right\}. \quad (142)$$

The budgets satisfy

$$\Gamma_1 \leq \Gamma_2 \leq \dots \leq \Gamma_T, \quad \Gamma_t - \Gamma_{t-1} \leq 1. \quad (143)$$

The scalar Γ_t is the modeler's conservativeness parameter. If $\Gamma_t = 0$, the uncertainty set collapses to the nominal forecast. If $\Gamma_t = t$, the full box uncertainty set is recovered.

The robust formulation requires each cost constraint in (138) to hold for every demand realization in U :

$$c_t \geq H \sum_{j \leq t} (x_j - D_j) \quad \forall D \in U, \quad c_t \geq B \sum_{j \leq t} (D_j - x_j) \quad \forall D \in U. \quad (144)$$

Substituting

$$D_j = \bar{D}_j + \hat{d}_j z_j \quad (145)$$

separates each constraint into a nominal part and a perturbation part. By symmetry of the uncertainty set, both worst cases are captured by the same protection function:

$$\Pi_t := \max_{z \in U} \sum_{j=1}^t \hat{d}_j z_j. \quad (146)$$

Thus the robust constraints can be written as

$$c_t \geq H \left(\sum_{j \leq t} (x_j - \bar{D}_j) + \Pi_t \right), \quad (147)$$

and

$$c_t \geq B \left(\sum_{j \leq t} (\bar{D}_j - x_j) + \Pi_t \right). \quad (148)$$

Since $\hat{d}_j \geq 0$, the inner problem defining Π_t becomes

$$\Pi_t = \max_z \left\{ \sum_{j \leq t} \hat{d}_j z_j : 0 \leq z_j \leq 1, \sum_{j \leq t} z_j \leq \Gamma_t \right\}. \quad (149)$$

Its LP dual is

$$\Pi_t = \min_{\alpha_t, \beta_{tj}} \left\{ \alpha_t \Gamma_t + \sum_{j \leq t} \beta_{tj} : \alpha_t + \beta_{tj} \geq \hat{d}_j, \forall j \leq t, \alpha_t, \beta_{tj} \geq 0 \right\}. \quad (150)$$

By strong duality, the primal and dual values are equal. Hence we can substitute the dual expression into the outer problem. The robust counterpart is the finite linear program

$$\begin{aligned} \min \quad & \sum_{t=1}^T (Px_t + c_t) \\ \text{s.t.} \quad & c_t \geq H \left(\sum_{j \leq t} (x_j - \bar{D}_j) + \alpha_t \Gamma_t + \sum_{j \leq t} \beta_{tj} \right), \quad \forall t, \\ & c_t \geq B \left(\sum_{j \leq t} (\bar{D}_j - x_j) + \alpha_t \Gamma_t + \sum_{j \leq t} \beta_{tj} \right), \quad \forall t, \\ & \alpha_t + \beta_{tj} \geq \hat{d}_j, \quad \forall j \leq t, \forall t, \\ & \alpha_t, \beta_{tj}, x_t \geq 0. \end{aligned} \quad (151)$$

Uncertainty has been converted into the protection price $\alpha_t \Gamma_t + \sum_{j \leq t} \beta_{tj}$, and α_t prices the uncertainty budget, while β_{tj} prices each individual deviation cap.

With setup costs, we add

$$\sum_{t=1}^T Fy_t \quad (152)$$

to the objective and impose

$$0 \leq x_t \leq My_t, \quad y_t \in \{0, 1\}. \quad (153)$$

The robustification is unchanged, because the uncertainty appears only in continuous-variable constraints.

The robust counterpart above is constraint-wise robust. Each constraint is protected against its own worst-case realization of z . The holding constraint effectively protects against persistently low demand, while the backlog constraint protects against persistently high demand. But nature chooses only one demand trajectory. It cannot choose different worst-case paths for different constraints in the same plan.

The more honest formulation is therefore the min-max problem

$$\min_{x \geq 0} \max_{z \in U} \sum_{t=1}^T \left(Px_t + \max \left\{ H \sum_{j \leq t} (x_j - D_j), B \sum_{j \leq t} (D_j - x_j) \right\} \right), \quad D_j = \bar{D}_j + \hat{d}_j z_j. \quad (154)$$

This formulation is less conservative because the adversary chooses one trajectory for all constraints. However, it is harder to solve, because the same z now couples all constraints.

Goes back to the constraints of (138), before any dualization, we can notice that the constraints contain raw D_t , not $\bar{D}_t + \alpha_t \Gamma_t + \sum \beta_{tj}$, and we can robustify the problem in a second, different way by

$$\min_{x \geq 0} \max_{z \in U} C(x, D(z)), \quad (155)$$

where $C(x, D)$ is the true cost of running plan x against trajectory D , evaluated by the epigraph variables c_t pressing down onto

$$\begin{aligned} & \max \left\{ H \sum_{j \leq t} (x_j - D_j), B \sum_{j \leq t} (D_j - x_j) \right\}. \quad (156) \\ & \min_x \max_{d, z} \sum_{t=1}^T (Px_t + c_t) \\ & \text{s.t.} \quad c_t \geq H \left(\sum_{j=1}^t (x_j - D_j) \right), \quad \forall t = 1, \dots, T, \\ & \quad c_t \geq B \left(\sum_{j=1}^t (-x_j + D_j) \right), \quad \forall t = 1, \dots, T, \quad (157) \\ & \quad D_t = \bar{D}_t + d_t z_t, \quad \forall t = 1, \dots, T, \\ & \quad -1 \leq z_t \leq 1, \quad \forall t = 1, \dots, T, \\ & \quad \sum_{j=1}^t |z_j| \leq \Gamma_t, \quad \forall t = 1, \dots, T, \\ & \quad x_t \geq 0, \quad \forall t = 1, \dots, T. \end{aligned}$$

One adversary. One trajectory. The same z must do all the damage everywhere at once. The new optimization problem solves the same reality problem with the same uncertainty set but a different answer to the question: Whom are we playing against? In the new model, conservatism is reduced, and honesty is restored.

The shared z couples all constraints, so the constraint-by-constraint dualization that produced the robust counterpart no longer applies: after eliminating c , the inner problem is the maximization of a convex

piecewise-linear function of z over the budget polytope $U = \{z : |z_t| \leq 1, \sum_{j \leq t} |z_j| \leq \Gamma_t\}$, which admits no single-LP reformulation. Reformulation having reached a dead end, the resolution is algorithmic—and the algorithm is obtained by reading (157) *twice*, once from each player’s point of view.

Fix the adversary’s choices to a finite set of demand scenarios $\mathcal{U} \subset U$ (the scenarios discovered so far) and optimize over x only. The universal quantification over U is replaced by replication of the cost constraints over $\mathcal{U}$:

$$(DMP) \quad \min_{x \geq 0} \sum_{t=1}^T (Px_t + c_t) \quad \text{s.t.} \quad c_t \geq H \sum_{j=1}^t (x_j - D_j^*), \quad c_t \geq B \sum_{j=1}^t (D_j^* - x_j), \quad \forall t, \forall D^* \in \mathcal{U}. \quad (158)$$

This is an ordinary LP ($\mathcal{U}$ finite, everything linear); at its optimum c_t equals the worst period- t cost over the scenarios in $\mathcal{U}$. Because $\mathcal{U} \subsetneq U$, (158) is a relaxation of (157). The plan is protected only against attacks already on record, hence its optimal value is a lower bound:

$$LB = v(DMP) \leq v^*, \quad v^* := \text{optimal value of (157)}. \quad (159)$$

Now fix the plan at the DMP optimizer x^* and optimize over the adversary’s variables only, searching the full set U :

$$(AP) \quad \max_{z \in U} \sum_{t=1}^T \left(Px_t^* + \max \left\{ H \sum_{j=1}^t (x_j^* - D_j), B \sum_{j=1}^t (D_j - x_j^*) \right\} \right), \quad D_j = \bar{D}_j + \hat{d}_j z_j. \quad (160)$$

Problem (160) maximizes a convex piecewise-linear function over the polytope U ; its optimum is therefore attained at an extreme point of U , and in implementation it is solved as a small MIP (binaries selecting the active branch of each max, plus linearization of $|z_j|$). Because x^* is feasible for (157) and (160) computes its exact worst-case cost, its value is an upper bound:

$$v^* \leq \max_{z \in U} C(x^*, D(z)) = UB. \quad (161)$$

Combining the two bounds,

$$\underbrace{v(DMP; \mathcal{U})}_{LB} \leq v^* \leq \underbrace{\max_{z \in U} C(x^*, D(z))}_{UB}, \quad (162)$$

which yields the iterative (scenario-generation) algorithm:

1. Solve (158) with the current $\mathcal{U} \Rightarrow$ plan x^* and LB.
2. Solve (160) at $x^* \Rightarrow$ worst-case scenario $\hat{d}$ and UB.
3. If $UB - LB \leq \varepsilon$, stop: (162) certifies that x^* is ε -optimal for (157). Otherwise set $\mathcal{U} \leftarrow \mathcal{U} \cup \{\hat{d}\}$ and return to Step 1.

Each round, the adversary exposes the specific weakness of the current plan, and the decision maker re-optimizes with that attack on record. Adding a scenario adds constraints to (158), so LB is nondecreasing; UB need not be monotone (each new plan has its own worst case), so one tracks the best upper bound found. Termination is finite: the AP only ever returns extreme points of the polytope U , of which there are finitely many, and once a vertex belongs to $\mathcal{U}$, it can no longer yield a strict separation (implying that no attack strategy can be effective repeatedly), and the library of attack strategies itself is finite.

Literature: Bertsimas and Thiele (2006); Bienstock and Özbay (2008); Zhang et al. (2025)

4. Bounds in Stochastic optimization

4.1 Bounding two-stage SP via functional & distributional approximation

Stochastic Linear Program:

$$\min_x \left\{ c^\top x + \mathbb{E}_P [Q(x, \xi)] : Ax = b \right\}. \quad (163)$$

where, for a given realization ξ of $\boldsymbol{\xi}$,

$$\begin{aligned} Q(x, \xi) := \min_y \quad & q^\top(\xi)y \\ \text{s.t.} \quad & Wy = h(\xi) - T(\xi)x, \\ & y \geq 0, \end{aligned} \quad (164)$$

with the decision sequence $x \rightarrow \xi \rightarrow y(\xi, x)$: commit to x , observe ξ , then take corrective action y .

The computational bottleneck is the evaluation, for a fixed candidate first-stage decision x_0 , of

$$Q(x_0) \equiv \mathbb{E}_P [Q(x_0, \xi)] \equiv \int_{\Xi} Q(x_0, \xi) P(d\xi). \quad (165)$$

This is a multidimensional integral of a function that is only implicitly defined. Each evaluation of the integrand requires solving an LP. Therefore, for most Q and most continuous distributions P , it cannot be computed exactly. If ξ has d independent components and each component is discretized into k points, exact evaluation requires k^d LP solves, which is exponential in the dimension. The classical response is to replace the hard object with an easier one, in one of two dual ways:

- **Functional approximation.** Keep the true distribution P , but replace the integrand $Q(x_0, \cdot)$ by simpler functions $L(\xi) \leq Q(x_0, \xi) \leq U(\xi)$ whose expectations under P are trivial to compute (typically because they only involve a few moments of P). Then $\mathbb{E}_P[L] \leq Q(x_0) \leq \mathbb{E}_P[U]$. (Jensen, 1906; Edmundson, 1957; Madansky, 1959; Wallace, 1987; Birge, 1988; Birge and Wets, 1989; Morton and Wood, 1999)
- **Distributional approximation.** Keep the true integrand $Q(x_0, \cdot)$, but replace P by an extremal discrete distribution Z (chosen from a set of distributions matching known information about P) such that $\mathbb{E}_Z[Q(x_0, \xi)]$ is computable, as a finite sum of LP values, and provably bounds $Q(x_0)$ from above or below. (Jensen, 1906; Edmundson, 1957; Madansky, 1959; Ben-Tal and Hochman, 1972; Dokov and Morton, 2005; Dulá, 1992; Edirisinghe, 1999; Edirisinghe and Ziemba, 1994, 1992; Frauendorfer, 1988; Edmundson, 1957; Kall, 1988)

Bounds based on distributional approximation: the generalized moment problem All known about P is its support Ξ and a finite list of generalized moments:

$$\mathbb{E}_P[g_i(\xi)] = m_i, \quad i \in I, \quad (166)$$

where the g_i are chosen moment functions. For example, $g_i(\xi) = \xi^i$ gives raw moments, while $g_0 = 1$ encodes total mass. Define the ambiguity class

$$\mathcal{P} := \{Z : Z(\Xi) = 1, \Xi \text{ convex compact}, \mathbb{E}_Z[g_i(\xi)] = m_i, \forall i \in I\}. \quad (167)$$

Since $P \in \mathcal{P}$, any Z solving

$$\text{Upper bound: } \max_{Z \in \mathcal{P}} \mathbb{E}_Z[f(\xi)], \quad \text{Lower bound: } \min_{Z \in \mathcal{P}} \mathbb{E}_Z[f(\xi)] \quad (168)$$

with $f(\xi) = Q(x_0, \xi)$ sandwiches the true value. Two facts make this practical:

- **Finite support of extremal measures:** The optimal Z^* is supported on at most $|I| + 1$ points of Ξ . Intuitively, the feasible set $\mathcal{P}$ is defined by $|I| + 1$ linear equality constraints on the space of measures, namely the $|I|$ moment constraints plus the normalization constraint. Its extreme points

are measures whose support cannot be split without violating a constraint. A dimension-counting, Richter–Rogosinski-type argument caps the number of atoms at the number of constraints. Since a linear functional

$$Z \mapsto \mathbb{E}_Z[f] \quad (169)$$

attains its optimum at an extreme point, the worst-case or best-case distribution is a small discrete measure. Therefore, evaluating the bound means solving only $|I| + 1$ recourse LPs.

- **Design freedom:** The entire literature on such bounds can be read as different choices of the moment functions g_i such that Z^* can be located analytically rather than by solving the moment problem numerically (Birge and Wets, 1987).

First-moment information on an interval gives Jensen and Edmundson-Madansky bounds. Adding second moments means conditioned on simplicial cells, and related information gives the family of refinements associated with Ben-Tal-Hochman, Frauendorfer, Gassmann-Ziemba, Edirisinghe-Ziemba, and Dokov-Morton.

Bounds based on Functional Approximation Restrict the approximating functions to the span of the same moment functions:

$$L(\xi) \text{ or } U(\xi) = u_0 + \sum_{i \in I} u_i g_i(\xi) \quad (170)$$

The payoff function is

$$\mathbb{E}_P[u_0 + \sum_i u_i g_i(\xi)] = u_0 + \sum_i u_i m_i \quad (171)$$

and the expectation is exactly computable from the known moments, regardless of how complicated P is. The best bounds of this form are semi-infinite LPs in the coefficient vector u :

$$\text{Lower bound:} \quad \max_u \left\{ u_0 + \sum_{i \in I} u_i m_i : L(\xi) \leq Q(x_0, \xi), \forall \xi \in \Xi \right\}. \quad (172)$$

$$\text{Upper bound:} \quad \min_u \left\{ u_0 + \sum_{i \in I} u_i m_i : U(\xi) \geq Q(x_0, \xi), \forall \xi \in \Xi \right\}. \quad (173)$$

Strong Duality Link: Distribution and function approximation are linked by strong duality.

Take the Lagrangian dual of the upper-bound problem as an example. Attach a nonnegative measure $Z(d\xi)$ to the constraint family

$$U(\xi) - Q(x_0, \xi) \geq 0. \quad (174)$$

Stationarity in u_0 forces

$$Z(\Xi) = 1, \quad (175)$$

and stationarity in u_i forces

$$\mathbb{E}_Z[g_i] = m_i. \quad (176)$$

What remains is exactly

$$\max_{Z \in \mathcal{P}} \mathbb{E}_Z[Q(x_0, \xi)]. \quad (177)$$

So:

$$\min_{U \geq Q(x_0, \cdot)} \mathbb{E}_P[U(\xi)] = \max_{Z \in \mathcal{P}} \mathbb{E}_Z[Q(x_0, \xi)]. \quad (178)$$

Symmetrically, for the lower bound, one obtains

$$\max_{L \leq Q(x_0, \cdot)} \mathbb{E}_P[L(\xi)] = \min_{Z \in \mathcal{P}} \mathbb{E}_Z[Q(x_0, \xi)]. \quad (179)$$

This is the same primal-dual pairing as in the moment-ambiguity DRO problem: the moment problem

over distributions is dual to a polynomial, or generalized moment, majorant/minorant reformulation.

Complementary slackness carries the geometric picture: the extremal measure Z^* sits exactly on the points where the approximating function touches $Q(x_0, \cdot)$. That is,

$$L(\xi) = Q(x_0, \xi) \quad \text{or} \quad U(\xi) = Q(x_0, \xi) \quad (180)$$

at the support points of Z^* . This contact-point structure is what allows Z^* to be written down by inspection in classical cases.

Example: Take $\xi \sim \text{Uniform}[0, 1]$, so that

$$\Xi = [0, 1], \quad \mathbb{E}[\xi] = \frac{1}{2}. \quad (181)$$

Let the objective be

$$f(\xi) = \xi^4, \quad (182)$$

and use moment functions

$$g_i(\xi) = \xi^i, \quad i \in \{0, 1, 2\}, \quad (183)$$

incrementally. The true value is $\mathbb{E}[\xi^4] = \frac{1}{5}$.

Lower bounds: $L_0 \leq L_1 \leq L_2$, and upper bounds $U_0 \geq U_1 \geq U_2$ are the best constant, affine, and quadratic minorants/majorants of ξ^4 . Each is dual to an extremal measure.

- $i = 0$, mass: The best constant minorant touches at $\xi = 0$, so

$$Z_0^{*,\text{low}} = \delta_{\{0\}}, \quad (184)$$

giving

$$L_0 = f(0) = 0. \quad (185)$$

The best constant majorant touches at $\xi = 1$, so

$$Z_0^{*,\text{up}} = \delta_{\{1\}}, \quad (186)$$

giving

$$U_0 = f(1) = 1. \quad (187)$$

- $i \leq 1$, mass and mean: The lower extremal measure is

$$Z_1^{*,\text{low}} = \delta_{\{1/2\}}. \quad (188)$$

This is Jensen's bound:

$$L_1 = f\left(\frac{1}{2}\right) = \frac{1}{16}. \quad (189)$$

The upper extremal measure puts mass on the endpoints $\{0, 1\}$ with weights matching the mean. Hence

$$Z_1^{*,\text{up}} = \frac{1}{2}\delta_{\{0\}} + \frac{1}{2}\delta_{\{1\}}. \quad (190)$$

This gives the Edmundson-Madansky bound:

$$U_1 = \frac{1}{2}f(0) + \frac{1}{2}f(1) = \frac{1}{2}. \quad (191)$$

- $i \leq 2$, second moment added: Since

$$\mathbb{E}[\xi^2] = \frac{1}{3}, \quad (192)$$

the extremal measures become two-point distributions. The lower extremal measure is supported on

$$\left\{0, \frac{2}{3}\right\}, \quad (193)$$

while the upper extremal measure is supported on

$$\left\{\frac{1}{3}, 1\right\}. \quad (194)$$

Thus the sandwich is tightened further.

The pattern is that each added moment constraint shrinks the ambiguity set $\mathcal{P}$. Therefore, the maximum decreases and the minimum increases:

$$L_0 \leq L_1 \leq L_2 \leq \mathbb{E}[\xi^4] \leq U_2 \leq U_1 \leq U_0. \quad (195)$$

This gives a monotone refinement chain in the information set I . The number of contact points, or atoms, grows with $|I|$, consistently with the finite support result for extremal measures.

Jensen gives the lower bound: If $f(\xi)$ is a convex function over the support of the random variable ξ , then

$$\mathbb{E}_P[f(\xi)] \geq f(\mathbb{E}_P[\xi]). \quad (196)$$

Edmundson–Madansky gives the upper bound:

Take a convex function $f(\xi)$ on a bounded support $[a, b]$. The $U(\xi) = r\xi + s$ is the chord connecting the endpoints $(a, f(a))$ and $(b, f(b))$:

$$f(\xi) \leq U(\xi), \quad \forall \xi \in [a, b], \quad (197)$$

$r = \frac{f(b)-f(a)}{b-a}$ is the slope, and s is the intercept chosen so that the line passes through both endpoints. Equivalently, writing ξ as a convex combination

$$\xi = \lambda a + (1 - \lambda)b, \quad \lambda = \frac{b - \xi}{b - a}, \quad (198)$$

the chord is

$$U(\xi) = \lambda f(a) + (1 - \lambda)f(b). \quad (199)$$

Because U is affine, its expectation depends only on $\mathbb{E}_P[\xi]$. Thus we do not need the full distribution P :

$$\mathbb{E}_P[f(\xi)] \leq \mathbb{E}_P[U(\xi)] = f(a)\frac{b - \mathbb{E}_P[\xi]}{b - a} + f(b)\frac{\mathbb{E}_P[\xi] - a}{b - a}. \quad (200)$$

Among all distributions on $[a, b]$ with a given mean, the extremal one that pushes all mass to the boundary maximizes $\mathbb{E}[f]$ when f is convex.

4.2 Quality of the expected value solution

With

$$z(x, \xi) = c^\top x + Q(x, \xi), \quad Ax = b, x \geq 0 \quad (201)$$

we define *Recourse Problem* (RP) and *Wait and See* (WS), and the difference is *Expected Value of Perfect Information* (EVPI),

$$RP = \min_x \mathbb{E}_P[z(x, \xi)], \quad WS = \mathbb{E}_P \left[\min_x z(x, \xi) \right], \quad EVPI = RP - WS \geq 0, \quad (202)$$

EVPI is the maximum amount a rational decision-maker would pay for a perfect forecast of ξ ; *Expected Value problem* (EV) and *Expected Result of using the EV Solution* (EVV), and the difference is *Value of*

Stochastic Solution (VSS),

$$EV = \min_x z(x, \bar{\xi}), \quad \bar{\xi} = \mathbb{E}_P[\xi], \quad EEV = \mathbb{E}_P[z(x(\bar{\xi}), \xi)], \quad VSS = EEV - RP \geq 0. \quad (203)$$

and the VSS is the cost of ignoring uncertainty when taking a decision.

The overall ordering (minimization) is

$$EV \leq WS \leq RP \leq EEV \quad (204)$$

The Maggioni–Wallace refinements A big VSS says the EV solution is bad; it doesn't say how. [Maggioni and Wallace \(2012\)](#) introduces two surgical tests, each obtained by re-solving the stochastic program with partial information inherited from $\bar{x}(\bar{\xi})$.

LUSS: testing the skeleton. Hypothesis test: The EV solution's structure, the set of variables it leaves at zero, is correct, even if its nonzero values are wrong. Let

$$J = \{j : \bar{x}_j(\bar{\xi}) = 0\} \quad (205)$$

be the zero-pattern, or the skeleton complement, of the EV solution. Solve the stochastic program restricted to that structure:

$$\hat{x} \in \arg \min_{x \in X} \mathbb{E}_P[z(x, \xi)] \quad \text{s.t.} \quad x_j = \bar{x}_j(\bar{\xi}) = 0, \quad j \in J. \quad (206)$$

That is, keep the same variables switched off, but let the stochastic program freely re-optimize all the variables that the EV solution activated. Define the Expected Skeleton Solution Value (ESSV) and the loss:

$$ESSV := \mathbb{E}_P[z(\hat{x}, \xi)], \quad LUSS := ESSV - RP. \quad (207)$$

Since fixing variables only shrinks the feasible set,

$$RP \leq ESSV \leq EEV. \quad (208)$$

The second inequality holds because $\bar{x}(\bar{\xi})$ itself satisfies the fixing constraints, so it is feasible for the restricted problem. Therefore,

$$0 \leq LUSS \leq VSS. \quad (209)$$

If $LUSS = 0$, or is small, the EV skeleton is a perfect structure for the stochastic problem. In this case, all of VSS comes from wrong values on the right variables. Practically, this is valuable: one can solve the large stochastic program only over the support identified by the cheap deterministic run. If $LUSS \approx VSS$, the structure itself is wrong. The deterministic model activates the wrong variables, and nothing short of the full stochastic program fixes it.

LUDS: testing upgradability.

Hypothesis test: The EV solution is a correct starting point that merely needs to be scaled up. It errs only by doing too little, not by doing the wrong things.

Solve the stochastic program constrained to dominate the EV solution componentwise:

$$\tilde{x} \in \arg \min_{x \in X} \mathbb{E}_P[z(x, \xi)] \quad \text{s.t.} \quad x \geq \bar{x}(\bar{\xi}). \quad (210)$$

Define the Expected Input Value and the loss:

$$EIV := \mathbb{E}_P[z(\tilde{x}, \xi)], \quad LUDS := EIV - RP, \quad 0 \leq LUDS \leq VSS. \quad (211)$$

If $LUDS = 0$, the EV solution is upgradeable. The true stochastic optimum sits weakly above it

componentwise, so the deterministic plan is a valid lower envelope. Nothing it committed to was a mistake; it only under-committed. This also has an algorithmic payoff: warm-starting or constraining the stochastic program with

$$x \geq \bar{x}(\bar{\xi}) \quad (212)$$

preserves optimality while shrinking the search space. If *LUDS* is large, some components of the EV solution are already too high. In this case, genuinely different trade-offs are needed.

4.3 Bounds for Multistage Stochastic Optimization

The nested formulation of a multistage stochastic program is

$$\min_{x_0 \in X_0(\xi_0)} c_0(x_0, \xi_0) + \mathbb{E}_{P_1} \left[\min_{x_1} c_1(x_1, \xi_1) + \mathbb{E}_{P_2} \left[\cdots + \mathbb{E}_{P_T} \left[\min_{x_T} c_T(x_T, \xi_T) \right] \right] \right].$$

The meaning: make the stage-0 decision $x_0 \rightarrow$ observe the random quantity $\xi_1 \rightarrow$ make $x_1 \rightarrow$ observe $\xi_2 \rightarrow \cdots$, alternating in this way. Decisions must satisfy *non-anticipativity*: x_t may depend only on the history observed so far, $\xi^t = (\xi_0, \dots, \xi_t)$.

If the random process has only finitely many realizations, the problem becomes a *scenario tree*: every root-to-leaf path is a scenario. The trouble is that the size of the tree explodes exponentially with the number of stages.

So the strategy shifts: instead of seeking an exact solution, cheaply compute a lower bound (a floor) and an upper bound (a ceiling), sandwiching the true optimal value $v^*(P)$ between them. If the gap between the two is only 2–3%, that is, for practical purposes, already knowing the answer.

There is one more key motivation. The Jensen / Edmundson–Madansky bounds from two-stage theory rely on convexity of the value function in ξ , and as soon as a multistage problem contains integer variables, convexity is destroyed. [Maggioni and Pflug \(2016\)](#) requires no convexity, no continuity, no linearity whatsoever—its principle is purely an information relaxation, so it continues to hold for mixed-integer multistage problems.

Lower bounds: Let x^* be the optimal policy of the original problem. Restricted to any group $\Xi_i^{(\ell)}$, it is a feasible policy for that group’s subproblem, hence

$$v(P_i^{(\ell)}) \leq \mathbb{E}_{P_i^{(\ell)}}[z(x^*, \xi)]. \quad (213)$$

Weight both sides by $\pi_i^{(\ell)}$ and sum, using the mixture identity $P = \sum_i \pi_i^{(\ell)} P_i^{(\ell)}$:

$$v_\ell^* := \sum_{i=1}^{m_\ell} \pi_i^{(\ell)} v(P_i^{(\ell)}) \leq \mathbb{E}_P[z(x^*, \xi)] = v^*(P), \quad (214)$$

feasibility + weighted averaging, with no property of the shape of any function used anywhere, this is exactly why it also holds for mixed-integer problems.

For groupings of different coarseness to form a monotone chain, the groupings at each level must satisfy:

Support conditions:

- (i) $\Xi_i^{(\ell)} = \bigcup_{\Xi_k^{(\ell-1)} \subseteq \Xi_i^{(\ell)}} \Xi_k^{(\ell-1)}$ - every group at level ℓ is a union of several groups at level $\ell - 1$ (nested structure);
- (ii) $\bigcup_{i=1}^{m_\ell} \Xi_i^{(\ell)} = \Xi$ - the groups at each level, taken together, must cover all scenarios (groups may share f scenarios; $f = 0$ means a disjoint partition).

Probability conditions:

- (iii) $P = \sum_{i=1}^{m_\ell} \pi_i^{(\ell)} P_i^{(\ell)}$, where $\sum_i \pi_i^{(\ell)} = 1$, $\pi_i^{(\ell)} \geq 0$ - the true distribution is a convex combination of the group distributions (within-level compatibility);

- (iv) each $P_i^{(\ell)}$ can in turn be written as a convex combination of the level- $(\ell - 1)$ distributions $P_k^{(\ell-1)}$ (between-level compatibility).

Under these four conditions, every level- ℓ subproblem is itself a stochastic program decomposable by the level- $(\ell - 1)$ grouping, and it yields the main theorem—a monotonically increasing chain of lower bounds:

$$v_1^* \leq v_2^* \leq \dots \leq v_{\ell-1}^* \leq v_\ell^* \leq \dots \leq v^*(P). \quad (215)$$

The bottom of the chain consists of singleton groups $(\{\omega_j\}, \delta_{\omega_j})$; the top of the chain is (Ξ, P) itself. This is an anytime algorithm: climb the chain, stop whenever time runs out, and whatever number is in hand is always a valid lower bound.

There are two concrete grouping schemes.

If $f = 0$, Disjoint Partition: cut Ξ into non-overlapping blocks; the group distribution is then the conditional distribution:

$$P_i^{(\ell)} = \begin{cases} \frac{1}{\pi_i^{(\ell)}} P(\omega_j) \delta_{\omega_j}, & \omega_j \in \Xi_i^{(\ell)}, \\ 0, & \text{otherwise,} \end{cases} \quad \pi_i^{(\ell)} = \sum_{\omega_j \in \Xi_i^{(\ell)}} P(\omega_j), \quad (216)$$

i.e., $P_i^{(\ell)} = P(\cdot \mid \Xi_i^{(\ell)})$, $\pi_i^{(\ell)} = P(\Xi_i^{(\ell)})$.

Beware of one pitfall: grouping is done on scenarios (complete root-to-leaf paths); each subproblem inherits only the subtree spanned by that group's scenarios. If scenarios sharing a long common history are grouped together, the subtree retains more branching structure $\rightarrow$ tighter bound; if unrelated scenarios are forced together, the subproblem nearly degenerates into a deterministic problem $\rightarrow$ the bound regresses toward WS.

If $f \geq 1$, Fixed Shared Scenario: designate a reference scenario ω_1 (the most typical path) and let it appear in every group: distribution:

$$(\Xi_1^{(2)}, \dots, \Xi_8^{(2)}) = (\{\omega_1, \omega_2\}, \{\omega_1, \omega_3\}, \dots, \{\omega_1, \omega_9\}). \quad (217)$$

Probabilities must be reweighted so that ω_1 keeps its original probability; the probability of each group's private scenarios is rescaled by

$$\frac{1 - P(\omega_1)}{\text{sum of the group's private probabilities}}; \quad (218)$$

the group weight is

$$\pi_i = \frac{\text{sum of the group's private probabilities}}{1 - P(\omega_1)}. \quad (219)$$

Every subproblem now contains a genuine comparison between the reference script and this group's own script, i.e., every subproblem truly confronts uncertainty, so for subproblems of the same size the bound is typically tighter; the cost is that the shared scenario gets solved repeatedly.

Upper bounds: The ceiling uses the opposite trick: take any concrete feasible policy, freeze it, and honestly compute its expected cost over the whole tree. The true performance of a feasible policy is necessarily $\geq$ the optimal value; hence, it is a valid upper bound. For instance, the deterministic problem along a single scenario ω , and then fix only its decisions for the first t stages, leaving all later stages free to be re-optimized:

$$UB_\omega^t := \text{fix the solution along scenario } \omega \text{ through stage } t, \text{ then solve the full problem}, \quad UB^t := \min_{\omega \in \Omega_T} UB_\omega^t.$$

Smaller t gives a tighter but more expensive bound ($t = 0$ is the original problem); larger t gives a

cheaper but looser bound.

A hand-designed chain (e.g., a naive bisection) carries no optimality guarantee whatsoever at the same group cardinality, different partitions can yield very different values of v_ℓ^* . The intuition: scenarios that are hard for any single common policy to serve well simultaneously should be placed in the same group (forcing them to coordinate is precisely the force that lifts the bound above WS); scenarios that any single policy can already serve well gain nothing from being grouped together.

Formalization: given a candidate solution set S and dual information λ from the non-anticipativity Lagrangian relaxation, define the score

$$w_{k,s}(\lambda) := P_k v_k(s) + \lambda^\top A_k s - v_k^*(\lambda), \quad s \in S, \omega_k \in \Xi_i^{(\ell)},$$

and the estimated improvement of group $\Xi_i^{(\ell)}$,

$$\vartheta_{\ell,i} := \min_{s \in S} \sum_{\omega_k \in \Xi_i^{(\ell)}} w_{k,s}(\lambda).$$

If some single candidate solution can serve the whole group well, ϑ is small (this grouping gains nothing); if no single candidate solution suits the whole group, ϑ is large (this grouping is valuable).

One then solves a small-scale MIP: maximize $\sum_\ell \sum_i \vartheta_{\ell,i}$, subject to constraints: assignment variables $y_{k,\ell,i} \in \{0,1\}$, fixed-scenario variables u_k ($\sum_k u_k = f$, fixed scenarios must enter every group), group-cardinality constraints, McCormick-type linking inequalities for pairwise co-grouping variables $v_{a,b,\ell,i}$, and, critically, the between-level consistency constraint

$$\sum_{i=1}^{m_\ell} v_{a,b,\ell,i} \geq \frac{1}{m_{\ell-1}} \sum_{j=1}^{m_{\ell-1}} v_{a,b,\ell-1,j},$$

which enforces the nesting condition and thereby guarantees that the optimized chain remains monotone. Numerically (on a multistage mixed-integer production problem), for the same subproblem budget, the optimized grouping significantly outperforms naive bisection.

4.4 Bounds via Stochastic Dominance

The discussion on bounds for multistage SP problem has two built-in restrictions. First, it assumes finitely many realizations: the random process is already a scenario tree, and the bounds tell you how to decompose that tree. But real models start from continuous distributions, and the tree is itself an approximation whose error the grouping bounds say nothing about. Second, the chain bounds the risk-neutral expectation functional; when the objective is risk-averse (CVaR-type, or a nested composition of risk measures), the average the group values step needs rethinking. [Maggioni and Pflug \(2019\)](#) addresses the problems: given a continuous (nondiscrete) multistage distribution P and a possibly risk-averse objective, construct finite tree processes $\tilde{P}$ (dominating) and $\underline{P}$ (dominated) such that

$$v^*(P_-) \leq v^*(P) \leq v^*(\tilde{P}), \tag{220}$$

so that solving two tractable tree problems certifiably sandwiches the intractable continuous one. The engine is stochastic dominance: an ordering on distributions that translates, through monotonicity or convexity of the cost function, into an ordering on optimal values.

Conceptually, this is based on Jensen/Edmundson–Madansky: we replace P by discrete measures ordered against it in a stochastic-dominance sense, which is the right notion when (a) the horizon is multistage, so the replacement must respect conditional structure, and (b) the functional is a risk measure, not an expectation.

Let $P, \tilde{P}$ be probability distributions on $\mathbb{R}^m$.

First-order stochastic dominance (FSD):

$$P \preceq_{FSD} \tilde{P} \quad (221)$$

if for every nondecreasing integrable function f ,

$$\mathbb{E}_P[f(\xi)] = \int_{\Xi} f(\xi) P(d\xi) \leq \int_{\Xi} f(\xi) \tilde{P}(d\xi) = \mathbb{E}_{\tilde{P}}[f(\xi)]. \quad (222)$$

$\tilde{P}$ puts its mass further to the right. For every threshold, $\tilde{P}$ assigns at least as much probability to exceeding it. Any decision-maker who prefers more to less prefers $\tilde{P}$. In one dimension, this is equivalent to pointwise ordering of CDFs. It also has a coupling interpretation by constructing

$$\xi \sim P, \quad \tilde{\xi} \sim \tilde{P}, \quad (223)$$

on the same probability space such that

$$\xi \leq \tilde{\xi} \quad \text{almost surely.} \quad (224)$$

Convex order (CXD):

$$P \preceq_{CXD} \tilde{P} \quad (225)$$

if the same inequality holds for every convex integrable function f :

$$\mathbb{E}_P[f(\xi)] \leq \mathbb{E}_{\tilde{P}}[f(\xi)]. \quad (226)$$

Example: If taking

$$f(\xi) = \pm \xi, \quad (227)$$

both convex forces equal means. If taking

$$f(\xi) = (\xi - c)^2 \quad (228)$$

shows that $\tilde{P}$ has more spread around every point. Hence $\tilde{P}$ is a mean-preserving spread of P : same center, more dispersion. Every risk-averse agent, with convex disutility, dislikes $\tilde{P}$ more.

The two orders serve different cost structures, and this is the organizing principle of the whole section:

- cost monotone in ξ , for example demand only ever raises cost, implies using FSD for both bounds;
- cost convex in ξ , for example a recourse LP with RHS randomness, implies using CXD for both bounds.

FSD in multistage A process $\xi = (\xi_1, \dots, \xi_T)$ is *totally monotonic* if its conditional distributions are monotone in the history:

$$\xi_t \mid (\xi_1 = u_1, \dots, \xi_{t-1} = u_{t-1}) \preceq_{FSD} \xi_t \mid (\xi_1 = w_1, \dots, \xi_{t-1} = w_{t-1}) \quad (229)$$

whenever

$$u_1 \leq w_1, \dots, u_{t-1} \leq w_{t-1}. \quad (230)$$

In words, a higher past makes a stochastically higher future. This is positive serial dependence in the FSD sense.

Let ξ and $\tilde{\xi}$ both be totally monotonic, and suppose there is stagewise conditional dominance along equal histories:

$$\xi_t \mid (\xi_1 = u_1, \dots, \xi_{t-1} = u_{t-1}) \preceq_{FSD} \tilde{\xi}_t \mid (\tilde{\xi}_1 = u_1, \dots, \tilde{\xi}_{t-1} = u_{t-1}). \quad (231)$$

Then the joint laws satisfy

$$P \preceq_{FSD} \tilde{P}. \quad (232)$$

Total monotonicity is needed because the natural proof couples the processes stage by stage. At stage t , the coupled trajectories have

$$\xi^{t-1} \leq \tilde{\xi}^{t-1} \quad (233)$$

componentwise, and one needs the conditional law of $\tilde{\xi}_t$ given the higher history $\tilde{\xi}^{t-1}$ to still dominate that of ξ_t given the lower history ξ^{t-1} . Stagewise dominance alone compares equal histories; total monotonicity bridges from equal histories to ordered histories. Chaining the two gives dominance along the whole trajectory, hence FSD of the joint law.

If the multistage cost

$$Q(x_0, \xi_1, x_1, \dots, \xi_T, x_T) \quad (234)$$

is monotone, namely nondecreasing in ξ , and the feasible-set dependence is compatible, meaning monotone in x , then the optimal value inherits the order:

$$P \preceq_{FSD} \tilde{P} \implies v^*(P) \leq v^*(\tilde{P}). \quad (235)$$

Heuristically, fix any policy. Along the coupling $\xi \leq \tilde{\xi}$, monotone costs give pathwise ordering of costs. Taking expectations preserves the inequality. Then optimize. Therefore, both value functions are monotone functionals of the driving law.

FSD example: Take $T = 3$ with two random stages (ξ_1, ξ_2) supported on a rectangle $\Xi \subset \mathbb{R}^2$.

- Step 1: grid the support. Cut the rectangle into cells

$$A_{i,j} = [a_i, a_{i+1}] \times [b_{i,j}, b_{i,j+1}], \quad (236)$$

where the ξ_1 -axis cuts $\{a_i\}$ are common, and the ξ_2 -cuts $\{b_{i,j}\}$ may depend on the ξ_1 -strip i . This freedom accommodates conditional heteroskedasticity: cells can be adapted to how $\xi_2 \mid \xi_1$ spreads in each strip.

Let

$$P_{i,j} := P(\cdot \mid A_{i,j}), \quad p_{i,j} := P(A_{i,j}). \quad (237)$$

Then

$$P = \sum_{i,j} p_{i,j} P_{i,j}. \quad (238)$$

This is the law of total probability.

- Step 2: build the upper tree $\tilde{P}$. Push the mass in each cell to the upper-right corner. Define a finite two-stage tree process $\tilde{\xi} = (\tilde{\xi}_1, \tilde{\xi}_2)$. Set

$$\tilde{\xi}_1 = a_{i+1} \quad \text{with probability} \quad \sum_j p_{i,j}. \quad (239)$$

Conditionally on $\tilde{\xi}_1 = a_{i+1}$, set

$$\tilde{\xi}_2 = b_{i,j+1} \quad \text{with probability} \quad \frac{p_{i,j}}{\sum_j p_{i,j}}. \quad (240)$$

Within each cell, all mass is pushed to the upper corner. Conditional on any history, $\tilde{\xi}_t$ sits at the upper edge of every cell containing mass of ξ_t . Hence each conditional law dominates in FSD. The trees inherit total monotonicity from the grid ordering, and the lemma then gives

$$P \preceq_{FSD} \tilde{P}. \quad (241)$$

- Step 3: build the lower tree $\underline{P}$. Push the mass in each cell to the lower-left corner. Symmetrically,

define

$$\underline{\xi}_1 = a_i, \quad \underline{\xi}_2 = b_{i,j} \quad (242)$$

with the same weights. Then

$$\underline{P} \preceq_{FSD} P. \quad (243)$$

- Step 4: the guaranteed sandwich. If

$$Q(x_0, \xi_1, x_1, \xi_2, x_2) \quad (244)$$

is monotone in (ξ_1, ξ_2) and in x , then

$$v^*(\tilde{P}) \geq v^*(P) \geq v^*(\underline{P}). \quad (245)$$

Both bounding problems are finite-tree multistage programs—the number of nodes equals the number of grid cells—solvable by any MILP/LP solver. Refining the grid tightens both sides (each refinement’s corner trees dominate/are dominated by the coarser ones), giving a two-sided refinement chain, and the chain converges toward the continuous-distribution value. Note what is not assumed: no convexity anywhere—monotonicity is a genuinely different structural hypothesis, and it survives integer variables.

CXD: generalized Jensen and Edmundson–Madansky When the cost is convex in ξ , as in the classical recourse situation, the right order is CXD. The two bounds are exactly the partition-refined versions of the classical pair.

If

$$P = \sum_i w_i P_i, \quad w_i \geq 0, \quad \sum_i w_i = 1, \quad (246)$$

then

$$\underline{P} := \sum_i w_i \delta_{\mathbb{E}(P_i)} \preceq_{CXD} P. \quad (247)$$

Proof in one line: for every convex function f , per-component Jensen gives

$$f(\mathbb{E}(P_i)) \leq \mathbb{E}_{P_i}[f]. \quad (248)$$

Weighting and summing yields

$$\mathbb{E}_{\underline{P}}[f] \leq \mathbb{E}_P[f]. \quad (249)$$

Geometrically, partition Ξ into cells A_i and collapse each cell’s mass to its barycenter $\mathbb{E}(P_i)$. For $Q(x_0, \xi_1, x_1)$ convex in ξ_1 , this gives

$$v^*(\underline{P}) \leq v^*(P). \quad (250)$$

With the trivial partition, namely one cell, this is Jensen’s bound and the EV problem. Refining the partition gives conditional Jensen, monotonically tightening the discrete measure. Thus $\underline{P}$ is a scenario tree whose nodes are cell barycenters. This is precisely how practitioners justify barycentric scenario generation as a guaranteed lower-bounding discretization.

Within each cell A_i , disintegrate every point $u \in A_i$ onto the extreme points $\{e_j^i\}$ of A_i . Write

$$u = \sum_j w_j^i(u) e_j^i, \quad (251)$$

where $w_j^i(u)$ are barycentric coordinates. Define

$$P_u^i := \sum_j w_j^i(u) \delta_{e_j^i}. \quad (252)$$

Then average over the cell:

$$\bar{P}_i := \int_{A_i} P_u^i dP(u), \quad \bar{P} := \sum_i \bar{P}_i. \quad (253)$$

The measure $\bar{P}$ is a probability measure and satisfies

$$P \preceq_{CXD} \bar{P}. \quad (254)$$

The mechanism is that replacing each point u by a mean-preserving lottery over the cell's corners preserves the mean but increases dispersion. This is an elementary dilation, and aggregating these dilations gives CXD dominance. Therefore, for convex recourse cost Q ,

$$v^*(P) \leq v^*(\bar{P}). \quad (255)$$

On a rectangle, the extremal points are the four corners, and the weights are products of one-dimensional chord weights. This is the Edmundson-Madansky formula applied cell by cell. In higher dimensions, the same tensorization applies, with 2^d corners per box.

What licenses doing this cell by cell are the following implication:

$$P_{i,j}^1 \preceq_{CXD} P_{i,j}^2 \quad \forall i, j, \quad P^k = \sum_{i,j} p_{i,j} P_{i,j}^k \implies P^1 \preceq_{CXD} P^2. \quad (256)$$

Convex order is preserved under mixtures. This follows immediately from the linearity of expectation in the measure. Therefore, local, or per-cell, dominance assembles into global dominance.

Example: Independent case: If $Q(x_0, \xi_1, x_1, \xi_2, x_2)$ is convex in (ξ_1, ξ_2) and $\xi_2 \perp \xi_1$, then

$$v^*(\bar{P}) \geq v^*(P) \geq v^*(\underline{P}). \quad (257)$$

Thus we obtain clean two-sided bounds: the corner-measure tree gives the upper bound, and the barycenter tree gives the lower bound.

Dependent case: With serial dependence, the barycenter construction has a subtlety. Collapsing ξ_1 to a cell barycenter perturbs the conditional law of ξ_2 that should be attached to it. In general, the conditional distribution at the barycenter is not the cell-average of the conditional distributions. Therefore, pure Jensen reasoning leaves a residual. The lower bound survives with an explicit correction term:

$$v^*(\bar{P}) \geq v^*(P) \geq v^*(\underline{P}) - \left[\sum_i p_i C(z_1^i) \right] \cdot \sup_i \left\| \bar{z}_1^i - z_1^{i,j} \right\|. \quad (258)$$

Structurally, the correction term is

$$\text{local Lipschitz constant} \times \text{cell diameter}. \quad (259)$$

More precisely, $C(\cdot)$ is a local Lipschitz-type constant of the value with respect to the stage-1 variable, and the distance term measures the gap between the cell barycenter $\bar{z}_1^i$ and the within-cell points $z_1^{i,j}$.

There are two readings. First, the correction vanishes as the grid is refined, because the supremum shrinks with the cell size. Hence the sandwich still converges. Second, the correction quantifies exactly the price of serial dependence in barycentric discretization. Under independence, this price is zero.

4.5 Bounds for Multistage Mixed-Integer DRO via Scenario Grouping

The same "split $\rightarrow$ solve $\rightarrow$ recombine" skeleton still survives when every expectation is replaced by a worst-case over an ambiguity set, but with a genuinely new difficulty: the ambiguity radius becomes a resource that must be budgeted between the subproblems and the recombination step, and the budgeting rule depends on which divergence defines the ambiguity set.

Multistage DRO in nested form:

$$\min_{x_0 \in \mathcal{X}_0(\xi_0)} c_0(x_0, \xi_0) + \max_{P_1 \in \mathcal{P}_1|\xi_0} \mathbb{E}_{P_1} \left[\min_{x_1} c_1(x_1, \xi_1) + \max_{P_2 \in \mathcal{P}_2|\xi_1} \mathbb{E}_{P_2} \left[\cdots + \max_{P_T \in \mathcal{P}_T|\xi_{T-1}} \mathbb{E}_{P_T} \left[\min_{x_T} c_T(x_T, \xi_T) \right] \right] \right]. \quad (260)$$

Each stage carries a conditional ambiguity set: a ball of distributions around the nominal conditional distribution $Q_{t|\xi^{t-1}}$, that is, the one estimated from data or from the scenario tree, with radius ρ_t :

$$\mathcal{P}_{t|\xi^{t-1}} := \{P_{t|\xi^{t-1}} \in \mathcal{M}(\Xi_{t|\xi^{t-1}}) : \Delta(P_{t|\xi^{t-1}}, Q_{t|\xi^{t-1}}) \leq \rho_t\}. \quad (261)$$

The model says the decision-maker trust the tree's conditional branching probabilities only up to distance ρ_t ; nature adversarially re-weights within that ball at every node. If $\rho_t = 0$, that is the risk-neutral problem. If $\rho_t \rightarrow \infty$, it is full robust optimization on the support. Mixed-integer decisions are allowed throughout. Nothing in the bounding argument requires convexity in x .

Two families of distance ϕ -divergences: for discrete distributions $P = \{p_{\omega_i}\}$, $Q = \{q_{\omega_i}\}$ on Ξ , define

$$\Delta_\phi(P, Q) := \sum_{i=1}^{|\Xi|} q_{\omega_i} \phi\left(\frac{p_{\omega_i}}{q_{\omega_i}}\right), \quad (262)$$

where ϕ is convex on $u \geq 0$ and satisfies

$$\phi(1) = 0. \quad (263)$$

Hence

$$\Delta_\phi(Q, Q) = 0. \quad (264)$$

We also use the usual conventions at zero.

Divergence	$\phi(u)$	$\Delta_\phi(P, Q)$
Variation distance	$ u - 1 $	$\sum_i p_{\omega_i} - q_{\omega_i} $
Cressie–Read power ($\theta \neq 0, 1$)	$\frac{1 - \theta + \theta u - u^\theta}{\theta(1 - \theta)}$	$\frac{1 - \sum_i p_{\omega_i}^\theta q_{\omega_i}^{1-\theta}}{\theta(1 - \theta)}$
J-divergence	$(u - 1) \log u$	$\sum_i (p_{\omega_i} - q_{\omega_i}) \log\left(\frac{p_{\omega_i}}{q_{\omega_i}}\right)$
χ -divergence of order $a > 1$	$ u - 1 ^a$	$\sum_i q_{\omega_i} \left \frac{p_{\omega_i}}{q_{\omega_i}} - 1 \right ^a$

(265)

ϕ -divergence balls only re-weight scenarios already in the support. Absolute continuity is built in: nature can inflate or deflate branch probabilities, but cannot invent new scenarios.

Wasserstein distance: the optimal-transport LP is

$$\begin{aligned} \Delta_W(P, Q) := \min_{z \geq 0} \quad & \sum_{i,j} d_{\omega_i, \omega_j} z_{\omega_i, \omega_j} \\ \text{s.t.} \quad & \sum_i z_{\omega_i, \omega_j} = q_{\omega_j}, \quad \forall j, \\ & \sum_j z_{\omega_i, \omega_j} = p_{\omega_i}, \quad \forall i. \end{aligned} \quad (266)$$

The ground cost is

$$d_{\omega_i, \omega_j} = \|\eta_{\omega_i} - \eta_{\omega_j}\|_\zeta, \quad \zeta \in \{1, 2, \infty\}, \quad (267)$$

which measures the distance between scenario values.

Unlike ϕ -divergences, Wasserstein balls are geometry-aware. Moving mass to a nearby scenario is cheap, while moving it to a distant scenario is expensive. This is why the group-distance construction will be needed.

A function $\mathcal{R} : L^\infty(\Xi, \mathcal{F}, Q) \rightarrow \mathbb{R}$ is a *coherent risk measure* if it satisfies convexity, monotonicity, translation equivariance, and positive homogeneity. Here, translation equivariance means

$$\mathcal{R}(\eta + \lambda) = \mathcal{R}(\eta) + \lambda. \quad (268)$$

The Artzner–Delbaen–Eber–Heath dual representation says that every coherent risk measure can be written as a worst-case expectation over some convex compact set of probability measures:

$$\mathcal{R}(\eta) = \max_{P \in \mathcal{P}} \mathbb{E}_P[\eta]. \quad (269)$$

Therefore,

$$\min_{x \in X} \mathcal{R}(c(x, \xi)) = \min_{x \in X} \max_{P \in \mathcal{P}} \mathbb{E}_P[c(x, \xi)]. \quad (270)$$

So a multistage DRO with nested ambiguity sets is a nested risk-averse program, and conversely. Concretely, CVaR_α corresponds to the ambiguity set

$$\left\{ P : p_i \leq \frac{q_i}{1 - \alpha} \right\}. \quad (271)$$

The variation-distance ball corresponds to a known mixture of expectation and CVaR.

For the dissection, given the nominal tree (Ξ, Q) , split the scenario set into groups

$$\Xi = \bigcup_{g=1}^{m_\ell} \Xi_g^{(\ell)}, \quad (272)$$

with nominal group measures $Q_g^{(\ell)}$ and weights

$$\pi_g^{(\ell)} \geq 0, \quad \sum_g \pi_g^{(\ell)} = 1, \quad Q = \sum_g \pi_g^{(\ell)} Q_g^{(\ell)}. \quad (273)$$

Let $\mathcal{G}$ be the σ -algebra generated by the groups. The procedure is as follows.

1. **Solve each group's DRO subproblem.** This is the same nested DRO restricted to $\Xi_g^{(\ell)}$, with inner radii $\bar{\rho}_{t,g}$, obtaining optimal values z_g^* .
2. **Recombine adversarially.** Instead of fixed-weight average, $\sum_g \pi_g z_g^*$, solve an outer DRO over the group values. Treat $\{z_g^*\}$ as a random variable on the m_ℓ groups, with nominal weights $\bar{Q} = \{\pi_g\}$, and with its own radius $\bar{\rho}$:

$$z^* := \max_{\tilde{P} \in \mathcal{P}_{\bar{\rho}}} \mathbb{E}_{\tilde{P}} \left[\{z_g^*\}_{g=1}^{m_\ell} \right]. \quad (274)$$

This is the first-level criterion, or single step recombination.

In the original problem, nature's re-weighting acts across groups. If we recombined with fixed weights π_g , then we would have granted the decision-maker extra power through group-tailored decisions, but denied nature her cross-group moves. These are two errors in opposite directions, and no bound is guaranteed. The construction restores nature's cross-group power in the outer step.

In the original problem, nature has a single budget ρ_t per stage for re-weighting all scenarios. In the decomposition, this power is split into within-group re-weighting, with budget $\bar{\rho}_{t,g}$ inside subproblem g , and across-group re-weighting, with budget $\bar{\rho}$ in the outer problem. The composition of these two moves must not exceed what the single budget ρ_t allowed. Otherwise, the decomposed adversary is stronger

than the true one, and the bound could overshoot.

The theorems below are precisely sub-additivity or composition conditions on the radii. They guarantee that nested balls with radii $(\bar{\rho}, \bar{\rho}_{\max})$ compose into something inside the original ball of radius ρ_1 . The exact shape of the condition depends on how each divergence behaves under mixture composition.

Theorem 4.1. First-level criteria

Let z^* be the optimal value of the multistage DRO with ϕ -divergence ambiguity sets of radii $\rho_1, \dots, \rho_T$. Group the scenarios, run the procedure above, and let

$$\bar{\rho}_{\max} = \max_g \bar{\rho}_{1,g}. \quad (275)$$

If the stage-1 radii satisfy the row of the following table corresponding to the divergence in use, then the decomposed value gives a valid lower bound.

Divergence	Radius condition
Variation distance	$\bar{\rho} + \bar{\rho}_{\max} + \bar{\rho} \bar{\rho}_{\max} \leq \rho_1$
J-divergence	$\bar{\rho} + \bar{\rho}_{\max} \leq \rho_1$
Cressie–Read, $\theta \in (0, 1)$	$\bar{\rho} + \bar{\rho}_{\max} \leq \rho_1$
Cressie–Read, $\theta < 0$ or $\theta > 1$	$\bar{\rho} + \bar{\rho}_{\max} - \theta(1 - \theta)\bar{\rho} \bar{\rho}_{\max} \leq \rho_1$
χ -divergence, order $a > 1$	$[(\bar{\rho})^{1/a} + (\bar{\rho}_{\max})^{1/a} + (\bar{\rho} \bar{\rho}_{\max})^{1/a}]^a \leq \rho_1$

The later stages are left untouched: $\bar{\rho}_{t,g} = \rho_t$, $t = 2, \dots, T$. Then

$$\underline{z}^* \leq z^*. \quad (277)$$

For J-divergence and Cressie–Read with $\theta \in (0, 1)$, the budget splits additively. That is, one can spend ρ_1 in any way between the inner and outer problems. Variation distance charges a positive cross-term, $\bar{\rho} \bar{\rho}_{\max}$, so the split is slightly wasteful. Cressie–Read outside the interval $(0, 1)$ has a negative cross-term, reflecting mild synergy. Finally, χ -divergence composes through a -th roots. These are triangle-inequality-like composition laws for each divergence under the mixture structure $P = \sum_g \pi_g P_g$.

Theorem 4.2. Multi-level criteria

Let z^* be the optimal value of a multistage DRO problem with ambiguity sets built via Wasserstein distance or any ϕ -divergence, with risk parameters $\rho_1, \rho_2, \dots, \rho_T$.

Consider the groups

$$\Omega_T = \bigcup_{g=1}^{m_\ell} \Omega_g^{(\ell)} \quad (278)$$

and the procedure described above. Let

$$\bar{\rho}_{t,\max} = \max_{g=1, \dots, m_\ell} \bar{\rho}_{t,g}, \quad t = 1, \dots, \tau. \quad (279)$$

If the following radius conditions hold:

Distance / divergence	Radius condition	
Wasserstein distance	$\bar{\bar{\rho}}_t + \bar{\rho}_{t,\max} \leq \rho_t$	
Variation distance	$\bar{\bar{\rho}}_t + \bar{\rho}_{t,\max} + \bar{\bar{\rho}}_t \bar{\rho}_{t,\max} \leq \rho_t$	
J-divergence	$\bar{\bar{\rho}}_t + \bar{\rho}_{t,\max} \leq \rho_t$	(280)
Cressie–Read power $\theta \in (0, 1)$	$\bar{\bar{\rho}}_t + \bar{\rho}_{t,\max} \leq \rho_t$	
Cressie–Read power $\theta < 0$ or $\theta > 1$	$\bar{\bar{\rho}}_t + \bar{\rho}_{t,\max} - \theta(1 - \theta)\bar{\bar{\rho}}_t \bar{\rho}_{t,\max} \leq \rho_t$	
χ -divergence of order $a > 1$	$[(\bar{\bar{\rho}}_t)^{1/a} + (\bar{\rho}_{t,\max})^{1/a} + (\bar{\bar{\rho}}_t \bar{\rho}_{t,\max})^{1/a}]^a \leq \rho_t$	

for $t = 1, \dots, \tau$, and if

$$\bar{\rho}_{t,g} = \rho_t, \quad t = \tau + 1, \dots, T, \quad (281)$$

then

$$\underline{z}^* \leq z^*. \quad (282)$$

For ϕ -divergences, the outer DRO over group values needs no geometry: re-weighting is just re-weighting. For Wasserstein distance, however, the outer ambiguity ball needs a ground distance between groups, and it must be chosen so that nature’s decomposed transport is no cheaper than her original transport. The paper’s definition is

$$d_{g_1, g_2} = \begin{cases} \max_{i \in \Omega_\tau^{(g_1)}, j \in \Omega_\tau^{(g_2)}} d_{i,j}, & g_1 \neq g_2, \\ 0, & g_1 = g_2. \end{cases} \quad (283)$$

This is the max-linkage, or worst-pair, distance between the groups’ scenario sets. It is computed among comparable nodes, meaning nodes sharing the same ancestor

$$a(n_1) = a(n_2), \quad (284)$$

so that the comparison respects the tree’s conditional structure. For example, one may have

$$d_{1,2} = \max \{ \|\xi_6 - \xi_8\|, \|\xi_6 - \xi_9\|, \|\xi_7 - \xi_8\|, \|\xi_7 - \xi_9\| \}. \quad (285)$$

Using the maximum makes cross-group transport in the outer problem at least as expensive as any true scenario-to-scenario move it stands in for. This is the conservative choice that keeps the decomposed adversary weaker, and hence keeps the bound valid. Singleton groups collapse to ordinary scenario distances, for example

$$d_{1,2} = \|\xi_2 - \xi_3\|. \quad (286)$$

Tractable reformulation of DRO A DRO problem contains a nested min-max the decision-maker minimizes, but inside sits nature’s adversarial maximization

$$\max_{P \in \mathcal{P}} \mathbb{E}_P[\eta] \quad (287)$$

over an ambiguity ball. You cannot hand a min-max problem to a standard solver. The technique is to dualize the inner max. Since nature’s problem is a convex optimization problem over probability weights, namely a linear objective and a convex ball constraint, strong duality replaces it by an equivalent minimization. This minimization then merges with the outer minimum into a single-level convex, LP, or MIP problem that CPLEX/Gurobi can solve directly.

The dual takes a different concrete form depending on the distance defining the ambiguity ball.

1. **Variation distance.** The worst case collapses to a closed-form mixture of a worst-scenario term

and a CVaR-type term:

$$\frac{\rho}{2} \max(\cdot) + \left(1 - \frac{\rho}{2}\right) \cdot \text{CVaR-like average.} \quad (288)$$

It can be implemented with a few auxiliary variables θ_n , $\bar{\omega}_n$, ψ_n , and linear constraints per node.

2. **Wasserstein distance.** LP duality of the optimal-transport problem, namely Kantorovich duality, yields dual variables γ_n for the ambiguity radius and $\nu_{n,j}$ together with linear constraints involving the scenario distances $d_{n,j}$. This is the Esfahani-Kuhn reformulation, applied nodewise on the scenario tree.

3. **General ϕ -divergence.** The dual is given by the conjugate-function formula:

$$\max_{P \in \mathcal{P}_\phi} \mathbb{E}_P[\eta] = \min_{\lambda \geq 0, \mu} \left\{ \mu + \rho\lambda + \lambda \mathbb{E}_Q \left[\phi^* \left(\frac{\eta - \mu}{\lambda} \right) \right] \right\}, \quad (289)$$

where ϕ^* is the convex conjugate of ϕ . Plugging in the conjugate of the chosen divergence, for example the piecewise $\phi_{m\chi^2}^*$ for modified χ^2 , gives a convex program with two extra variables per node.

In the multistage setting, this dualization is applied at every node of the scenario tree. Each node carries its own conditional ambiguity ball, so the nested min-max-min-max structure telescopes into one large single-level model with auxiliary variables attached to each node.

Upper Bounds UB_ω^t : solving the full problem by inserting, until stage t , the optimal solution of the problem under scenario $\omega \in \Omega_T$.

$$UB^t := \min_{\omega \in \Omega_T} \{UB_\omega^t\}. \quad (290)$$

How to Split the Ambiguity Budget ($\bar{\rho}$ vs. $\bar{\rho}$) There are the practical design rules for allocating the original radius ρ between the inner, or within-group, ambiguity budget $\bar{\rho}_{\max}$ and the outer, or across-group, ambiguity budget $\bar{\rho}$, subject to the composition conditions of Theorems 1-2. For example, in the additive case, the condition is

$$\bar{\rho} + \bar{\rho}_{\max} \leq \rho. \quad (291)$$

Observation 1: the tree is indexed by the group cardinality l .

- **Top: one big group, l large.** All scenarios are kept in one subproblem. Nature should therefore receive her full power inside the subproblem:

$$\bar{\rho} = 0, \quad \bar{\rho}_{\max} = \rho. \quad (292)$$

More scenarios per subproblem give sharper lower bounds.

- **Middle: moderate groups.** As l decreases, the ambiguity budget should be shifted outward:

$$\bar{\rho} \uparrow, \quad \bar{\rho}_{\max} \downarrow. \quad (293)$$

- **Bottom: singleton groups, $l = 1$.** Each subproblem contains a single deterministic scenario. There is nothing inside the group for nature to re-weight, so any inner budget is wasted. The best split is

$$\bar{\rho}_{\max} = 0, \quad \bar{\rho} = \rho. \quad (294)$$

All adversarial power is moved to the outer recombination step.

The ambiguity budget should sit where the scenarios are: inside large groups, and outside small groups.

Observation 2: which recombination criteria to use.

- **Small group cardinality l .** Multi-level criteria work better. Nesting outer DRO steps across several levels re-introduces adversarial coupling at multiple resolutions, compensating for very small subproblems.

- **Large group cardinality l .** First-level, or single-step, criteria work better. The subproblems already carry much of the coupling, so extra nesting only fragments the ambiguity budget.

Group size determines both where the radius goes and how to recombine. The radius moves monotonically from inner to outer as groups become smaller, and the recombination rule moves from single-step to multi-level as the dissection becomes finer. The singleton limit is the clean boundary case:

$$(\bar{\rho}_{\max}, \bar{\rho}) = (0, \rho). \quad (295)$$

4.6 Bounds for Multi-Horizon Stochastic Optimization

Two uncertainty types:

- strategic: investment costs, carbon prices, etc;
- operational: load levels, hydro inflows, etc.

Standard multistage SP forces all uncertainty onto one scenario tree. However, with the above two kinds of uncertainty, the tree would need to branch at every hour, and strategic nodes would multiply the whole operational structure combinatorially. The multi-horizon formulation proposed by Kaut et al. (2014) makes a structural compromise: a coarse strategic tree with nodes $n \in \mathcal{N}_t$, and, hanging off each strategic node, a bundle of embedded operational scenarios $s \in \mathcal{S}_t$ with operational periods $\tau \in \mathcal{T}_t$. The key modeling assumption: operational outcomes at nodes n do not feed back into the strategic branching, and the two layers are linked only through the strategic decision variables (installed capacity constrains dispatch). This is what keeps the problem's size additive rather than multiplicative.

$$\text{MHSP} := \min_{\mathbf{x}, \mathbf{y}} \sum_{t=1}^H \sum_{n \in \mathcal{N}_t} \pi_n \left(c_n x_n + \sum_{s \in \mathcal{S}_t} w_n^s \sum_{\tau \in \mathcal{T}_t} q_n^{s, \tau} y_n^{s, \tau} \right) \quad (296)$$

Subject to

$$Ax_n = h_n, \quad n \in \mathcal{N}_1, \quad (297)$$

and the strategic dynamics linking a node to its parent,

$$T_n x_{a(n)} + W_n x_n = h_n, \quad n \in \mathcal{N}_t, t \in \mathcal{H} \setminus \{1\}, \quad (298)$$

a coupling constraint injecting the strategic state into operations,

$$T_n^{s,1} x_n + W_n^{s,1} y_n^{s,1} = h_n^{s,1}, \quad n \in \mathcal{N}_t, s \in \mathcal{S}_t, \quad n \in \mathcal{N}_t, \tau \in \mathcal{T}_t \setminus \{1\}, s \in \mathcal{S}_t, t \in \mathcal{H}, t \in \mathcal{H} \quad (299)$$

and the operational dynamics within each scenario,

$$T_n^{s, \tau} y_n^{s, \tau-1} + W_n^{s, \tau} y_n^{s, \tau} = h_n^{s, \tau}, \quad (300)$$

also

$$\begin{aligned} x_n &\in \mathbb{R}_+^{n_t}, \quad n \in \mathcal{N}_t, \quad t \in \mathcal{H}, \\ y_n^{s, \tau} &\in \mathbb{R}_+^{n_t^\tau}, \quad n \in \mathcal{N}_t, \quad \tau \in \mathcal{T}_t, \quad s \in \mathcal{S}_t, \quad t \in \mathcal{H}. \end{aligned} \quad (301)$$

Here, π_n denotes the probability of strategic node n , while w_n^s denotes the weight of operational scenario s . Even with multi-horizon compression, realistic instances may grow extremely large.

With two uncertainty layers, every classical construction (replaced by expectation; given perfect information) can be applied selectively to one layer or both, producing a richer taxonomy.

- **MHEV:** replace both strategic and operational uncertainty by expectations, one deterministic path with average parameters $\bar{c}_t$, $\bar{q}_t^\tau$, $\bar{h}_t$. This is the most straightforward instrument: the multi-horizon EV problem.
- **MHOEV:** keep the strategic tree complete, and replace only each node's operational scenarios by their expectation $\bar{q}_n^\tau$. This gives a half-deterministic problem, and it is still a tree, but with one

average operational profile per node.

- **MHWS:** grant perfect information on both layers, and solve one deterministic problem for each pair consisting of a strategic scenario s and an operational realization, and then average the resulting values. This is the full wait-and-see problem.
- **MHSWS:** the subtle case. The grant perfect information only on the strategic scenarios, while preserving operational non-anticipativity. Within each strategic scenario, the operational decisions must still satisfy $y_{s,t}^{s',\tau} = y_{s,t}^{s'',\tau}$, for operational scenarios s' and s'' that coincide up to period τ . Thus, the decision maker is clairvoyant about the investment environment, but not about the weather.

Lower bounds by ordering:

$$\text{MHWS} \leq \text{MHSWS} \leq \text{MHSP}, \quad (302)$$

and, under fixed objective coefficients and a fixed recourse matrix,

$$\text{MHEV} \leq \text{MHWS}, \quad \text{MHEV} \leq \text{MHOEV}. \quad (303)$$

Each inequality follows one argument: more information or more tailoring can only lower the minimum, while expectation substitution provides lower bounds through Jensen's inequality under the stated fixity conditions.

Upper bounds (fix-and-evaluate chains)

The idea is the stage-indexed EEV approach. The quantity MHEES_t fixes the strategic solution of the expected-value problem up to stage t and re-optimizes the remaining decisions. Similarly, MHEESO_t^τ additionally fixes the operational decisions up to operational period τ .

The corresponding chains are

$$\text{MHSP} \leq \text{MHEES}_1 \leq \text{MHEES}_2 \leq \dots \leq \text{MHEES}_{H-1}, \quad (304)$$

$$\text{MHSP} \leq \text{MHEESO}_1^\tau \leq \dots, \quad (305)$$

and

$$\text{MHEES}_t \leq \text{MHEESO}_t^\tau. \quad (306)$$

More fixing produces a computationally cheaper but looser bound. The bound may even be infinite, since fixing an expected-value decision can make the resulting problem infeasible under uncertainty.

The gap semantics:

The pairwise differences correspond to multi-horizon EVPIs and VSSs, and localize the value of information by layer:

- $$\text{MHSP} - \text{MHSWS}, \quad (307)$$

which measures the value of perfectly forecasting the strategic parameters;

- $$\text{MHSWS} - \text{MHWS}, \quad (308)$$

which measures the additional value of also forecasting the operational parameters;

- $$\text{MHEES}_t - \text{MHSP}, \quad (309)$$

which measures the cost of trusting the deterministic model's investment decisions through stage t .

Upper Bounds for Multi-Horizon SO : Partial Fixing and Rolling Horizon.

Partial Fixing

- MHEES_t : the donor is the solution of the EV problem. It also serves as a diagnostic, since the gap

$$\text{MHEES}_t - \text{MHSP} \quad (310)$$

measures the cost of trusting the deterministic investment plan through stage t , in the same spirit as the VSS family.

- MHES_t : the donor is the solution of a better but still solvable auxiliary problem, with further details provided in the unpublished paper (Micheli et al., 2025). Its purpose is purely computational: to obtain a ceiling that is as tight as computationally affordable for instances that cannot be solved directly.

Three steps:

1. Solve a cheap auxiliary problem to obtain a solution $\tilde{x}$. The auxiliary problem may be:
 - the deterministic expected-value problem MHEV, which gives the MHEES_t chain; or
 - a solvable reduced or grouped subproblem, which gives MHES_t .
2. Fix only its strategic decisions up to stage t :

$$x_n = \tilde{x}_n \quad (311)$$

for all strategic nodes n belonging to stages $t' \leq t$.

3. Re-optimize everything else on the full MHSP: strategic decisions after stage t and all operational decisions y remain completely free.

The resulting optimal value is an upper bound. Indeed, the partially fixed solution remains feasible for the original problem, provided that the fixing does not induce infeasibility. Therefore, its objective value satisfies

$$\text{partially fixed value} \geq \text{MHSP}. \quad (312)$$

$$\text{MHSP} \leq \text{MHEES}_1 \leq \text{MHEES}_2 \leq \dots \leq \text{MHEES}_{H-1}. \quad (313)$$

Rolling Horizon: The weak spot of fixing methods: the donor's decisions are produced once, offline, and after fixing, they never respond to new information. Rolling horizon adopts a different philosophy, imitating a real decision-maker: don't solve the whole horizon at once; at each stage, look at a limited window of the future through a telescope, decide, freeze, move forward, and look again. Real system operators (grid dispatchers and inventory managers) operate exactly this way, so the procedure naturally produces a feasible policy, and hence naturally yields an upper bound.

At each step, the method solves a *truncated subproblem* whose future horizon is divided into two segments:

- $\hat{e}$: *number of non-relaxed stages*: the next $\hat{e}$ strategic stages are modeled at full resolution, including complete strategic branching and the full bundle of operational scenarios at each node.
- $\hat{e}_R$: *number of relaxed stages*: the following $\hat{e}_R$ stages are modeled in subsampled or relaxed form. The operational scenarios in these stages are not all included but are sampled with selection probability ϕ_e . In the relaxed segment, each node randomly retains half of its operational scenarios.
- The more distant future is truncated entirely and is not considered in the current subproblem.

The algorithm:

1. At the current strategic node, construct the rolling-window subproblem consisting of $\hat{e}$ stages modeled at full resolution followed by $\hat{e}_R$ stages modeled in sampled or relaxed form, and solve it.
2. Freeze only the decisions of the current stage. All other decisions computed inside the window are discarded. They serve only as temporary scaffolding that makes the current decision more forward-looking.
3. Roll the window forward by one stage and repeat the procedure along every branch of the strategic scenario tree.

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4. Once the entire tree has been traversed, the frozen decisions form a complete feasible policy. Evaluate this policy on the original MHSP to obtain an upper bound.

The parameters

$$(\hat{e}, \hat{e}_R, \phi_e) \tag{314}$$

act as a regulator across three dimensions: quality, computation, and time.

- Larger $\hat{e}$: each step sees further and more clearly $\rightarrow$ better decisions, tighter bound, but larger subproblems per step;
- $\hat{e}_R, \phi_e$: a cheap middle setting between 'don't look at the farther future at all' ($\hat{e}_R = 0$, myopic) and 'look at it in full resolution' (equivalent to enlarging $\hat{e}$, expensive), a sampled, blurry sketch of what lies ahead reminds the current decision that 'more is coming,' avoiding myopia without paying full price;

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